Numerical analysis of superconducting phases in the extended Hubbard model with non-local pairing

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Abstract

[To be continued...]

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List of symbols and abbreviations

AF	Antiferromagnetic
BCS	Bardeen-Cooper-Schrieffer (theory)
β	Inverse temperature
$\Delta^{(\gamma)}$	γ-wave component of the gap function
DoF	Degree(s) of freedom
EHM	Extended Hubbard model
EV(s)	Expectation value(s)
FS	Fermi surface
HF	Hartree-Fock
HFP(s)	Hartree-Fock parameter(s)
HM	Hubbard model
L_ℓ	Number of lattice sites in direction ℓ
LRT	Linear Response Theory
MFT	Mean-Field Theory
NN(s)	Nearest neighbor(s)
RWC(s)	Relevant Wick's contraction(s)
RMP(s)	Renormalized model parameter(s)
SC	Superconductor
SDW	Spin-density wave
SSB	Spontaneous symmetry breaking
T_c	Critical temperature

Chapter 1

The normal phase

This chapter is devoted to the MFT analysis of the "normal phase", which is, the one where interacting electrons do not develop an antiferromagnetic or superconducting gap, although undergoing an effective non-trivial transformation. What will be derived here, in particular the hopping renormalization effect, constitutes a peculiarity of effective MFT treatment of the EHM and will be the groundwork of following analyses. Note that this chapter is structured in such a way that what is derived here mathematically is recovered and expanded to more complex phases in next chapters.

1.1 The "normal" phase and its symmetries

The normal phase is simply the one where no symmetry is broken, and the many-body ground state preserves the entire U(2) structure of Eq. (**??**) as well as translational symmetry. Thus, the normal phase is essentially given by the null gap limit of both the antiferromagnetic and superconducting phases. By discussing it separately now, we highlight a MFT feature of the EHM of particular interest, namely, the insurgence of a new, self-consistent hopping parameter. From Tabs. **??** and **??**, we know that the relevant Wick's contractions (RWCs) for this phase are

$\hat{H}_U$	Hartree contraction
$\hat{H}_V$ (o.s. sector)	Hartree contraction
$\hat{H}_V$ (s.s. sector)	Hartree and Fock contractions

and since the phase we are establishing obeys translational symmetry, we also know Hartree terms are essentially chemical potential shifts. Next section derives this effect in detail.

In the normal phase, the state retains its full $\mathrm{SU}(2) \times \mathrm{U}^{\mathrm{c}}(1) \times$ (Lattice) symmetry discussed in Sec. **??**. Spin is a pure physical degeneracy, being th e$\uparrow$ and $\downarrow$ sectors separated, and $\hat{H}$ is $\mathbb{Z}_2$ invariant with respect to spin inversion, $\sigma \leftrightarrow \overline{\sigma}$. This implies that all physical quantities, as is for the normal phase, should not depend on the spin index. As for the normal phase, we will perform all computation leaving σ explicit and using this consideration at the end.

1.2 Hartree effects: chemical potential renormalization

The first RMP we discuss is μ, whose numerical value is renormalized by Hartree channel interactions within MFT. In the context of numerical simulations at fixed density, the chemical potential is determined self-consistently; thus any net effect that shifts μ is effectively ignored within (and limitedly to) the iterative algorithm. However, the same is not true when we consider the free energy density, which – as will be discussed – takes a non-null contribution from μ itself. Hartree contractions originate a chemical potential shift, which is, the system itself contributes to μ with HFPs and thus allows the "physical" value of μ to be different than the effective one, eventually lowering the contribution to free energy.

1.2.1 Hartree channel in the normal phase: analytical derivation

Within this section we discuss analytically the effects of MFT on the Hartree channel, separating the local and non-local interactions and obtaining a set of HFPs needed to setup an interative HF algorithm.

Local repulsion. In the normal phase, the local repulsion $\hat{H}_U$ only contributes through Hartree-like RWCs,

$$\hat{H}_U \simeq U \sum_i \left[\langle \hat{n}_{i\uparrow}\rangle \hat{n}_{i\downarrow} + \hat{n}_{i\uparrow}\langle \hat{n}_{i\downarrow}\rangle - \langle \hat{n}_{i\downarrow}\rangle\langle \hat{n}_{i\uparrow}\rangle\right]$$

For a translationally invariant phase such as the normal phase, it holds

$$\langle \hat{n}_{i\sigma}\rangle = n \tag{1.1}$$

which gives immediately

$$\begin{aligned}\hat{H}_U &\simeq nU \sum_i \left[\hat{n}_{i\uparrow} + \hat{n}_{i\downarrow}\right] - U\sum_i n^2 \\ &= nU \times \hat{N} - E^{(\mathrm{N})}_{\mathrm{H}/U}\end{aligned}$$

being $\hat{N}$ the total number operator and $E^{(\mathrm{N})}_{\mathrm{H}/U}$ the "contraction energy" (also known as double counting term) due to the Hartree (H) contraction of the U repulsion. The chemical potential is corrected by

$$\mu \to \mu - nU \tag{1.2}$$

The non-local attraction further corrects μ.

Non-local attraction. Let us break down $\hat{H}_V$ isolating its Hartree RWCs in both o.s. and s.s. sectors:

$$\begin{aligned}\hat{H}_V \simeq\ & \overbrace{-V\sum_{\langle ij\rangle}\sum_{\sigma}\left[\langle \hat{n}_{i\sigma}\rangle \hat{n}_{j\sigma} + \hat{n}_{i\sigma}\langle \hat{n}_{j\sigma}\rangle - \langle \hat{n}_{i\sigma}\rangle\langle \hat{n}_{j\sigma}\rangle\right]}^{\text{s.s.}} \\ & \underbrace{-V\sum_{\langle ij\rangle}\sum_{\sigma}\left[\langle \hat{n}_{i\sigma}\rangle \hat{n}_{j\overline{\sigma}} + \hat{n}_{i\sigma}\langle \hat{n}_{j\overline{\sigma}}\rangle - \langle \hat{n}_{i\sigma}\rangle\langle \hat{n}_{j\overline{\sigma}}\rangle\right]}_{\text{o.s.}} + (\text{all the rest})\end{aligned} \tag{1.3}$$

where "all the rest" collects all non-Hartree RWCs. Recalling the Ansatz of Eq. (1.1), we get

$$\hat{H}_V \simeq \underbrace{-nV\sum_{\langle ij\rangle}\sum_{\sigma}\left[\hat{n}_{j\sigma} + \hat{n}_{i\sigma}\right]}_{\text{s.s.}} \underbrace{-nV\sum_{\langle ij\rangle}\sum_{\sigma}\left[\hat{n}_{j\overline{\sigma}} + \hat{n}_{i\sigma}\right]}_{\text{o.s.}} - E^{(\mathrm{N})}_{\mathrm{H}/V} + (\text{all the rest})$$

with $E^{(\mathrm{N})}_{\mathrm{H}/V}$ the normal state shift to energy brought by $\hat{H}_V$. Now, evidently both sums above are just a number operator,

$$\hat{H}_V \simeq -2nzV \times \hat{N} + (\text{all the rest})$$

This accounts for the final Hartree shift of μ,

$$\mu \to \mu + 2znV \tag{1.4}$$

Thus, when considering the net shift to μ due to both interactions combining Eqns. (1.2), (1.4), we get

$$\tilde{\mu} \equiv \mu + n(2zV - U) \tag{1.5}$$

This result remains valid in all phases discussed in this text: for the AF phase, the SDW character leaves this shift untouched, while the superconducting phase we are discussion is inherently translational invariant.

1.2.2 Hartree contraction energy

The double counting terms accounting for energy shifts in the Hartree channel are, as anticipated:

$$E^{(\mathrm{N})}_{\mathrm{H}/U} = \sum_i \langle \hat{n}_{i\downarrow}\rangle\langle \hat{n}_{i\uparrow}\rangle \tag{1.6}$$

$$= L_x L_y \times nU \tag{1.7}$$

for the local repulsion and

$$E^{(\mathrm{N})}_{\mathrm{H}/V} = V\sum_{\langle ij\rangle}\sum_{\sigma}\left[\langle \hat{n}_{i\sigma}\rangle\langle \hat{n}_{j\sigma}\rangle + \langle \hat{n}_{i\sigma}\rangle\langle \hat{n}_{j\overline{\sigma}}\rangle\right] \tag{1.8}$$

$$= 2V\sum_{\langle ij\rangle}\sum_{\sigma} n^2 \tag{1.9}$$

$$= L_x L_y \times 2zn^2 V \tag{1.10}$$

for the non-local attraction, being $z = 4$ the coordination number for the 2D square lattice. In the last passage, we used that

$$\sum_{\langle ij\rangle}(1) = \sum_{i\in\mathcal{S}/2}\sum_{j\in\mathrm{NN}(i)}(1) = \frac{L_x L_y}{2}\times z$$

as well as spin degeneracy. Note that, correctly, the contraction energy is an extensive quantity. This implies that we can define a contraction energy per site – something we will do later when computing total free energy density.

1.3 Fock effects: bare bands renormalization

Up to now, no relevant changes have been made to the starting situation of the conventional degenerate Fermi gas. The scenario changes now, with the analysis of the Fock RWCs. Even if no gap opens, in the normal phase the particles interact by the means of a non-local attraction, a feature that effectively redefines the bare fermionic bands.

From Wick's decomposition of $\hat{H}_V$, the only allowed Fock term comes from the same-spin part due to SU(2) symmetry selection rules. Let us focus only on this term when decomposing $\hat{H}$:

$$\hat{H}_V \simeq V\sum_{\langle ij\rangle}\sum_{\sigma}\left[\langle \hat{c}^\dagger_{i\sigma}\hat{c}_{j\sigma}\rangle\hat{c}^\dagger_{j\sigma}\hat{c}_{i\sigma} + \hat{c}^\dagger_{i\sigma}\hat{c}_{j\sigma}\langle \hat{c}^\dagger_{j\sigma}\hat{c}_{i\sigma}\rangle - \langle \hat{c}^\dagger_{i\sigma}\hat{c}_{j\sigma}\rangle\langle \hat{c}^\dagger_{j\sigma}\hat{c}_{i\sigma}\rangle\right] + (\text{all the rest}) \tag{1.11}$$

(note the + sign in front of the displayed term). A bond-wise hopping amplitude can be defined,

$$\tilde{t}_{ij\sigma} \equiv t - V\langle \hat{c}^\dagger_{j\sigma}\hat{c}_{i\sigma}\rangle$$

The effective diffusive hamiltonian is given by

$$\hat{H}_{\tilde{t}} = \hat{H}_t + V\sum_{\langle ij\rangle}\sum_{\sigma}\left[\langle \hat{c}^\dagger_{i\sigma}\hat{c}_{j\sigma}\rangle\hat{c}^\dagger_{j\sigma}\hat{c}_{i\sigma} + \mathrm{h.c.}\right]$$

$$= -\sum_{\langle ij\rangle}\sum_{\sigma}\left[\tilde{t}_{ij\sigma}\hat{c}^\dagger_{i\sigma}\hat{c}_{j\sigma} + \mathrm{h.c.}\right]$$

To proceed, it is useful to move to reciprocal space.

1.3.1 Fock channel in the normal phase: analytical derivation

In reciprocal space, the effective hopping must be transformed as well. Consider the Fourier Transform $\mathcal{F}$ given in Eq. (**??**), applied to the displayed part of Eq. (1.11),

$$V\sum_{\langle ij\rangle}\sum_{\sigma}\left[\langle \hat{c}^\dagger_{i\sigma}\hat{c}_{j\sigma}\rangle\hat{c}^\dagger_{j\sigma}\hat{c}_{i\sigma} + \hat{c}^\dagger_{i\sigma}\hat{c}_{j\sigma}\langle \hat{c}^\dagger_{j\sigma}\hat{c}_{i\sigma}\rangle - \langle \hat{c}^\dagger_{i\sigma}\hat{c}_{j\sigma}\rangle\langle \hat{c}^\dagger_{j\sigma}\hat{c}_{i\sigma}\rangle\right]$$

$$\overset{\mathcal{F}}{=} \frac{2V}{L_x L_y}\sum_{\mathbf{K},\mathbf{k},\mathbf{k}'}\sum_{\sigma}\left[\cos(\delta k_x) + \cos(\delta k_y)\right]\langle \hat{c}^\dagger_{\mathbf{K}+\mathbf{k}\sigma}\hat{c}_{\mathbf{K}-\mathbf{k}'\sigma}\rangle\hat{c}^\dagger_{\mathbf{K}-\mathbf{k}\sigma}\hat{c}_{\mathbf{K}+\mathbf{k}'\sigma} - E^{(\mathrm{N})}_{\mathrm{F}/V} \tag{1.12}$$

Here, the 2 prefactor comes from recognizing that the first two operators in square brackets in the first line generate identical contributions to the full sum; the double counting energy shift is given by

$$E^{(\mathrm{N})}_{\mathrm{F}/V} \equiv \frac{V}{L_x L_y} \sum_{\mathbf{K},\mathbf{k},\mathbf{k}'} \sum_{\sigma} \left[\cos(\delta k_x) + \cos(\delta k_y)\right] \langle \hat{c}^\dagger_{\mathbf{K}+\mathbf{k}\sigma} \hat{c}_{\mathbf{K}-\mathbf{k}'\sigma}\rangle \langle \hat{c}^\dagger_{\mathbf{K}-\mathbf{k}\sigma} \hat{c}_{\mathbf{K}+\mathbf{k}'\sigma}\rangle \tag{1.13}$$

Up to this point, this derivation is general and will be re-used in next chapters. Here the specificity of the physical phase settles in: if we assume the normal state to be a free degenerate Fermi gas with renormalized bands, we get

$$\langle \hat{c}_{\mathbf{k}_1\sigma_1} \hat{c}_{\mathbf{k}_2\sigma_2}\rangle = \delta_{\mathbf{k}_1=\mathbf{k}_2} \delta_{\sigma_1=\sigma_2} f(\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu})$$

being f the Fermi-Dirac distribution,

$$f(\epsilon; \beta, \mu) \equiv \frac{1}{e^{\beta(\epsilon-\mu)} + 1}$$

It follows that Eq. (1.12) becomes

$$\begin{aligned} V \sum_{\langle ij\rangle} \sum_{\sigma} & \left[\langle \hat{c}^\dagger_{i\sigma} \hat{c}_{j\sigma}\rangle \hat{c}^\dagger_{j\sigma} \hat{c}_{i\sigma} + \hat{c}^\dagger_{i\sigma} \hat{c}_{j\sigma} \langle \hat{c}^\dagger_{j\sigma} \hat{c}_{i\sigma}\rangle - \langle \hat{c}^\dagger_{i\sigma} \hat{c}_{j\sigma}\rangle \langle \hat{c}^\dagger_{j\sigma} \hat{c}_{i\sigma}\rangle\right] \\ & \overset{\mathcal{F}}{=} \frac{2V}{L_x L_y} \sum_{\mathbf{K},\mathbf{k}} \sum_{\sigma} \left[\cos(2k_x) + \cos(2k_y)\right] \langle \hat{c}^\dagger_{\mathbf{K}+\mathbf{k}\sigma} \hat{c}_{\mathbf{K}+\mathbf{k}\sigma}\rangle \hat{c}^\dagger_{\mathbf{K}-\mathbf{k}\sigma} \hat{c}_{\mathbf{K}-\mathbf{k}\sigma} - E^{(\mathrm{N})}_{\mathrm{F}/V} \end{aligned} \tag{1.14}$$

since the only contribution comes from $\mathbf{k}' = -\mathbf{k}$. Let now $\mathbf{q} \equiv \mathbf{K} + \mathbf{k}$, $\mathbf{q}' \equiv \mathbf{K} - \mathbf{k}$. Being for $\ell = x, y$

$$\begin{aligned} \delta q_\ell &\equiv q_\ell - q'_\ell \\ &= (K_\ell + k_\ell) - (K_\ell - k_\ell) \\ &= 2k_\ell \end{aligned}$$

we reduce Eq. (1.12) to

$$\begin{aligned} V \sum_{\langle ij\rangle} \sum_{\sigma} & \left[\langle \hat{c}^\dagger_{i\sigma} \hat{c}_{j\sigma}\rangle \hat{c}^\dagger_{j\sigma} \hat{c}_{i\sigma} + \hat{c}^\dagger_{i\sigma} \hat{c}_{j\sigma} \langle \hat{c}^\dagger_{j\sigma} \hat{c}_{i\sigma}\rangle - \langle \hat{c}^\dagger_{i\sigma} \hat{c}_{j\sigma}\rangle \langle \hat{c}^\dagger_{j\sigma} \hat{c}_{i\sigma}\rangle\right] \\ & \overset{\mathcal{F}}{=} \frac{2V}{L_x L_y} \sum_{\mathbf{q},\mathbf{q}'} \sum_{\sigma} \left[\cos(\delta q_x) + \cos(\delta q_y)\right] \langle \hat{c}^\dagger_{\mathbf{q}\sigma} \hat{c}_{\mathbf{q}\sigma}\rangle \hat{c}^\dagger_{\mathbf{q}'\sigma} \hat{c}_{\mathbf{q}'\sigma} - E^{(\mathrm{N})}_{\mathrm{F}/V} \end{aligned} \tag{1.15}$$

Recall now the result of Eq. (**??**),

$$\cos(\delta q_x) + \cos(\delta q_y) = \frac{1}{2} \sum_{\gamma} \varphi^{(\gamma)}_{\mathbf{q}} \varphi^{(\gamma)*}_{\mathbf{q}'} \quad \text{for } \gamma \in \{s^*, p_x, p_y, d_{x^2-y^2}\}$$

Thanks to this handy property, we reduce Eq. (1.15) to

$$\begin{aligned} V \sum_{\langle ij\rangle} \sum_{\sigma} & \left[\langle \hat{c}^\dagger_{i\sigma} \hat{c}_{j\sigma}\rangle \hat{c}^\dagger_{j\sigma} \hat{c}_{i\sigma} + \hat{c}^\dagger_{i\sigma} \hat{c}_{j\sigma} \langle \hat{c}^\dagger_{j\sigma} \hat{c}_{i\sigma}\rangle - \langle \hat{c}^\dagger_{i\sigma} \hat{c}_{j\sigma}\rangle \langle \hat{c}^\dagger_{j\sigma} \hat{c}_{i\sigma}\rangle\right] \\ & \overset{\mathcal{F}}{=} \sum_{\gamma} \left[\frac{1}{L_x L_y} \sum_{\mathbf{q}\in\mathrm{BZ}} \varphi^{(\gamma)}_{\mathbf{q}} u_{\mathbf{q}\sigma}\right] \times V \sum_{\mathbf{q}'\in\mathrm{BZ}} \varphi^{(\gamma)*}_{\mathbf{q}'} \hat{c}^\dagger_{\mathbf{q}'\sigma} \hat{c}_{\mathbf{q}'\sigma} - E^{(\mathrm{N})}_{\mathrm{F}/V} \end{aligned} \tag{1.16}$$

where we defined the EV:

$$u_{\mathbf{q}\sigma} \equiv \langle \hat{c}^\dagger_{\mathbf{q}\sigma} \hat{c}_{\mathbf{q}\sigma}\rangle \tag{1.17}$$

Of course, the numerical value of $u_{\mathbf{q}\sigma}$ is given by the Fermi-Dirac distribution; however, for the sake of reproducibility of this derivation in next chapters, we omit that for a moment. Let $u^{(\gamma)}_\sigma$ be its γ-wave component,

$$u^{(\gamma)}_\sigma = \frac{1}{L_x L_y} \sum_{\mathbf{q}\in\mathrm{BZ}} \varphi^{(\gamma)}_{\mathbf{q}} \langle \hat{c}^\dagger_{\mathbf{q}\sigma} \hat{c}_{\mathbf{q}\sigma}\rangle$$

Operator	Sector	RWCs	Net effect
$\hat{H}_U$		Hartree	μ shift by $-nU$
$\hat{H}_V$	o.s.	Hartree	μ shift by nzV
$\hat{H}_V$	s.s.	Hartree Fock	μ shift by nzV Bands renormalization and resymmetrization

Table 1.1 | Relevant Wick's contractions for the normal phase.

Now, the bare bands $\epsilon_{\mathbf{k}}$ are s^*-wave symmetric, but in general $\tilde{\epsilon}_{\mathbf{k}}$ can exhibit a more complex symmetry. Surely, due to spatial inversion invariance, it must be

$$u_{\mathbf{q}\sigma} = u_{-\mathbf{q}\sigma}$$

thus

$$u_\sigma^{(p_\ell)} = 0 \quad \text{for} \quad \ell \in \{x, y\}$$

recalling the antisymmetric form factors of Tab **??**. However, $u_\sigma^{(s^*)}$ and $u_\sigma^{(d)}$ are in general non-zero. Eq. (1.16) becomes

$$V \sum_{\langle ij \rangle} \sum_\sigma \left[\langle \hat{c}_{i\sigma}^\dagger \hat{c}_{j\sigma} \rangle \hat{c}_{j\sigma}^\dagger \hat{c}_{i\sigma} + \hat{c}_{i\sigma}^\dagger \hat{c}_{j\sigma} \langle \hat{c}_{j\sigma}^\dagger \hat{c}_{i\sigma} \rangle - \langle \hat{c}_{i\sigma}^\dagger \hat{c}_{j\sigma} \rangle \langle \hat{c}_{j\sigma}^\dagger \hat{c}_{i\sigma} \rangle \right]$$
$$\overset{\mathcal{F}}{=} V \sum_{\mathbf{q} \in \mathrm{BZ}} \sum_\sigma \left[u_\sigma^{(s^*)} \varphi_{\mathbf{q}}^{(s^*)*} + u_\sigma^{(d)} \varphi_{\mathbf{q}}^{(d)*} \right] \hat{c}_{\mathbf{q}\sigma}^\dagger \hat{c}_{\mathbf{q}\sigma} - E_{\mathrm{F}/V}^{(\mathrm{N})} \tag{1.18}$$

Let

$$\tilde{\epsilon}_{\mathbf{k}\sigma} \equiv \epsilon_{\mathbf{k}\sigma} + u_\sigma^{(s^*)} V (\cos q_x + \cos q_y) + u_\sigma^{(d)} V (\cos q_x - \cos q_y) \tag{1.19}$$

This equation defines the bands renormalization. The fact that both a s^* and a d-wave component appear introduces an interesting anisotropy that is discusses in next section. Tab. 1.1 lists the various RWCs and their net effect on the MFT hamiltonian.

1.3.2 Fock contraction energy

Finally, let us derive the Fock contraction energy by expanding Eq. (1.13):

$$\begin{aligned} E_{\mathrm{F}/V}^{(\mathrm{N})} &= \frac{V}{L_x L_y} \sum_{\mathbf{q},\mathbf{q}'} \sum_\sigma \left[\cos(\delta q_x) + \cos(\delta q_y) \right] \langle \hat{c}_{\mathbf{q}\sigma}^\dagger \hat{c}_{\mathbf{q}'\sigma} \rangle \langle \hat{c}_{\mathbf{q}\sigma}^\dagger \hat{c}_{\mathbf{q}'\sigma} \rangle \\ &= L_x L_y \times V \sum_{\gamma,\sigma} \left[\frac{1}{L_x L_y} \sum_{\mathbf{q} \in \mathrm{BZ}} \varphi_{\mathbf{q}}^{(\gamma)} u_{\mathbf{q}\sigma} \right] \left[\frac{1}{L_x L_y} \sum_{\mathbf{q}' \in \mathrm{BZ}} \varphi_{\mathbf{q}}^{\prime(\gamma)*} u_{\mathbf{q}'\sigma} \right] \\ &= L_x L_y \times V \sum_\sigma \left[\left| u_\sigma^{(s^*)} \right|^2 + \left| u_\sigma^{(d)} \right|^2 \right] \end{aligned} \tag{1.20}$$

This expression gives the Fock contraction energy coming from the $\hat{H}_V$ operator. The particularly relevant result, here, is the fact that the full spectral amplitude of $u_{\mathbf{k}\sigma}$ here is weighted and contributes to contraction energy with a V linear prefactor. This suggests that a mixed-symmetry band could be energetically preferred.

1.3.3 Extending hopping to d-wave bands

From Tabs. **??** and **??** we can inverse Fourier transform the form factors to get the full effective hopping,

$$\tilde{t}_{ij\sigma} = \sum_{\gamma \in \{s^*, s\}} t_\sigma^{(\gamma)} \varphi_{ij}^{(\gamma)}$$

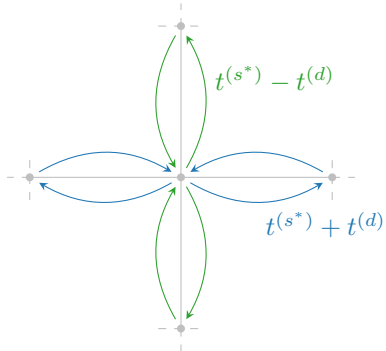


Figure 1.1 | Graphic rendition of anisotropic hopping, with a s^*-wave component (symmetric by $\pi/$ rotations) and a d-wave component (antisymmetric).

with

$$t_\sigma^{(s^*)} = t - \frac{u_\sigma^{(s^*)} V}{2} \qquad t_\sigma^{(d)} = -\frac{u_\sigma^{(d)} V}{2}$$

Interestingly enough, in principle an anisotropy in the normal diffusion between x and y direction emerges. [To be continued...]

1.4 Normal free energy density

An important quantity to derive in order to compare different phases is free energy density, defined as

$$f_{\mathrm{MFT}}^{(\mathrm{N})} \equiv \frac{F_{\mathrm{MFT}}^{(\mathrm{N})}}{L_x L_y}$$

being F_{MFT} the total free energy. Consider the complete hamiltonian

$$\hat{H} \simeq \sum_{\mathbf{k},\sigma} \tilde{\epsilon}_{\mathbf{k}\sigma} \hat{c}_{\mathbf{k}\sigma}^\dagger \hat{c}_{\mathbf{k}\sigma} + E_{\mathrm{c}}^{(\mathrm{N})} - \tilde{\mu}\hat{N}$$

where $E_{\mathrm{c}}^{(\mathrm{N})}$ is the (negative) total contraction energy for the normal phase,

$$E_{\mathrm{c}}^{(\mathrm{N})} \equiv -\left[E_{\mathrm{H}/U}^{(\mathrm{N})} + E_{\mathrm{H}/V}^{(\mathrm{N})} + E_{\mathrm{F}/V}^{(\mathrm{N})}\right]$$

having included the various contraction energies from Eqns. (1.7), (1.10) and (1.20).

For a free Fermi gas, the free energy density is simply obtained by summing four contributions:

1. The free particles thermal ground state energy density,

$$e_{\mathrm{g}}^{(\mathrm{N})} = \frac{2}{L_x L_y} \sum_{\mathbf{k}\in\mathrm{BZ}} \tilde{\epsilon}_{\mathbf{k}} f(\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu})$$

2. The contraction energy density,

$$\begin{aligned} e_{\mathrm{c}}^{(\mathrm{N})} &\equiv \frac{E_{\mathrm{c}}^{(\mathrm{N})}}{L_x L_y} \\ &= -nU - 2V\left[zn^2 + \left|u_\sigma^{(s^*)}\right|^2 + \left|u_\sigma^{(d)}\right|^2\right] \end{aligned}$$

3. The entropic free energy

$$e_s^{(\mathrm{N})} \equiv -Ts$$

with $T = 1/k_B\beta$ and s the free Fermi gas entropy density,

$$s = -\frac{2k_B}{L_x L_y} \sum_{\mathbf{k}\in\mathrm{BZ}} \left[(1 - f(\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu})) \ln(1 - f(\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu})) + f(\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu}) \ln f(\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu})\right]$$

We use the identity:

$$\ln\left(1-\frac{1}{e^x+1}\right)=x+\ln\left(\frac{1}{e^x+1}\right)$$

and reduce the above expression to:

$$s=-\frac{2k_B}{L_xL_y}\sum_{\mathbf{k}\in\mathrm{BZ}}\Big[\ln\left(1-f(\tilde{\epsilon}_{\mathbf{k}};\beta,\tilde{\mu})\right)+f(\tilde{\epsilon}_{\mathbf{k}};\beta,\tilde{\mu})\ln f(\tilde{\epsilon}_{\mathbf{k}};\beta,\tilde{\mu})$$
$$-\beta\tilde{\epsilon}_{\mathbf{k}}f(\tilde{\epsilon}_{\mathbf{k}};\beta,\tilde{\mu})-f(\tilde{\epsilon}_{\mathbf{k}};\beta,\tilde{\mu})\ln f(\tilde{\epsilon}_{\mathbf{k}};\beta,\tilde{\mu})\Big]$$

Second and fourth terms cancel out, and we have

$$e_s^{(\mathrm{N})}=\frac{2}{\beta L_xL_y}\sum_{\mathbf{k}\in\mathrm{BZ}}\ln\left(1-f(\tilde{\epsilon}_{\mathbf{k}};\beta,\tilde{\mu})\right)-\frac{2}{L_xL_y}\sum_{\mathbf{k}\in\mathrm{BZ}}\tilde{\epsilon}_{\mathbf{k}}f(\tilde{\epsilon}_{\mathbf{k}};\beta,\tilde{\mu})$$

4. The chemical potential correction

$$e_n^{(\mathrm{N})}=\mu n$$

Note here that we are using the "physical" value of μ, not its RMP counterpart $\tilde{\mu}$. The reason for that is the following: by studying a fixed density situation, $n=N/L_xL_y$, we are actually diagonalizing the hamiltonian operator

$$\hat{H}-\mu(\hat{N}-N)$$

by the means of the Lagrange multplier μ. This gives the entire energy a $+\mu N$ shifts which is not absorbed by the renormalization of μ. [Double check.]

Thus, composing everything

$$\begin{aligned}f_{\mathrm{MFT}}^{(\mathrm{N})}&\equiv e_{\mathrm{g}}^{(\mathrm{N})}+e_{\mathrm{c}}^{(\mathrm{N})}+e_s^{(\mathrm{N})}+e_n^{(\mathrm{N})}\\&=\frac{2}{\beta L_xL_y}\sum_{\mathbf{k}\in\mathrm{BZ}}\ln\left(1-f(\tilde{\epsilon}_{\mathbf{k}};\beta,\tilde{\mu})\right)-nU-2V\left[zn^2+\left|u_\sigma^{(s^*)}\right|^2+\left|u_\sigma^{(d)}\right|^2\right]+\mu n\end{aligned}\tag{1.21}$$

1.5 Hartree-Fock iterative algorithm

[To be continued...]

Chapter 2

Anti-Ferromagnetic instability

This chapter is devoted to an in-depth discussion of the antiferromagnetic (AF) long-range ordering in the EHM by the means of MFT. In the context of the 2D Hubbard model (HM), discussed in App. **??**, AF ordering establishes as an unconventional metallic phase which is only stable at half-filling (as is discussed in Sec. **??**). The situation for the EHM is similar: the present analysis is carried out based on the argument that, given the symmetry structure of the non-local NN interaction, the fundamental features of the commensurate AF phase of the HM are effectively preserved, while the parameters are redefined by the interaction itself. The two most relevant effects we discuss in this chapter are the "gap complexification" (Sec. [Missing]) and the "hopping renormalization" (Sec. [Missing]), the latter being a general feature of the EHM common to all phases preserving (or, at most, weakening without destroying) the translational invariance, discussed in Sec. 1.3.

2.1 The AF phase and its symmetries

When discussing the AF phase on the EHM square lattice, we must specify the type of antiferromagnetism we are dealing with. SDW antiferromagnets on a square lattice can emerge in a variety of ways, the simplest one – described in this text – being the so-called "commensurate". By this we mean essentially a kind of long range spin density ordering varying on the length scale of two lattice sites. This type of AF structure is particularly interesting on a square lattice because the phase transition is simply described in reciprocal space by the order parameter

$$\langle \hat{c}^\dagger_{\mathbf{k}\sigma} \hat{c}_{\mathbf{k}+\boldsymbol{\pi}\sigma} \rangle$$

going from a null EV in the normal phase to a non-zero value. This phenomenon occurs at the so called AF vector

$$\boldsymbol{\pi} \equiv (\pi, \pi)$$

Large portions of a square lattice Fermi surface (FS) can turn out to be connected precisely by an AF vector: this implies that, if the hamiltonian makes available a scattering channel at this wavevector, electrons can jump from one side to the other, making said EV non-zero.

In principle, the AF phase is physically established by breaking the SU(2) spin rotation symmetry down to the $\mathrm{U}^z(1)$ projected spin rotation symmetry; however, it is to be noted that – even if the concept of SSB is well defined only for the $\mathrm{SU}(2) \to \mathrm{U}^z(1)$ symmetry reduction – the hamiltonian in thermodynamic limit remains $\mathbb{Z}_2$ invariant with respect to spin inversion, $\sigma \leftrightarrow \overline{\sigma}$. Identical considerations as in Sec. 1.1 hold This: all physical quantities are spin-independent. As for the normal phase, we will use this considerations at the end of the calculations.

2.1.1 Antiferromagnetism in the HM

Within the Hubbard model, the antiferromagnetic phase (AF) is specified by the Ansatz (**??**) which is explicitly breaking translational invariance in each spin sector, while preserving $\mathrm{U}^z(1)$ and $\mathrm{U}^c(1)$ symmetries and reduces the hamiltonian to the form of Eq. (**??**) (ignoring the double-counting

EV	Meaning	Equation
$\langle \hat{s}^x_{\mathbf{k}\sigma} \rangle$	$\left\langle \hat{c}^\dagger_{\mathbf{k}\sigma} \hat{c}_{\mathbf{k}+\boldsymbol{\pi}\sigma} + \hat{c}^\dagger_{\mathbf{k}+\boldsymbol{\pi}\sigma} \hat{c}_{\mathbf{k}\sigma} \right\rangle$	(??)
$i \langle \hat{s}^y_{\mathbf{k}\sigma} \rangle$	$\left\langle \hat{c}^\dagger_{\mathbf{k}\sigma} \hat{c}_{\mathbf{k}+\boldsymbol{\pi}\sigma} - \hat{c}^\dagger_{\mathbf{k}+\boldsymbol{\pi}\sigma} \hat{c}_{\mathbf{k}\sigma} \right\rangle$	(??)
$\langle \hat{s}^z_{\mathbf{k}\sigma} \rangle$	$\left\langle \hat{c}^\dagger_{\mathbf{k}\sigma} \hat{c}_{\mathbf{k}\sigma} - \hat{c}^\dagger_{\mathbf{k}+\boldsymbol{\pi}\sigma} \hat{c}_{\mathbf{k}+\boldsymbol{\pi}\sigma} \right\rangle$	(??)

Table 2.1 List of the various expectation values (EVs) of each pseudospin component, their meaning in terms of fermionic operators and the reference equations from the pure HM.

terms)

$$\hat{H}_t + \hat{H}_U \overset{\text{MFT}}{\simeq} -t \sum_{\langle \mathbf{r}\mathbf{r}' \rangle} \sum_{\sigma} \left[\hat{c}^\dagger_{\mathbf{r}\sigma} \hat{c}_{\mathbf{r}'\sigma} + \text{h.c.} \right] + nU \sum_{\mathbf{r}} \left[\hat{n}_{\mathbf{r}\uparrow} + \hat{n}_{\mathbf{r}\downarrow} \right] - mU \sum_{\mathbf{r}} (-1)^{x+y} \left[\hat{n}_{\mathbf{r}\uparrow} - \hat{n}_{\mathbf{r}\downarrow} \right]$$

In reciprocal space, the hamiltonian decomposes as in Eq. (??),

$$\hat{H}_t + \hat{H}_U \overset{\text{MFT}}{\simeq} \sum_{\mathbf{k}\in\text{MBZ}} \sum_{\sigma} \hat{\Psi}^\dagger_{\mathbf{k}\sigma} h_{\mathbf{k}\sigma} \hat{\Psi}_{\mathbf{k}\sigma} \quad \text{being} \quad h_{\mathbf{k}\sigma} \equiv \begin{bmatrix} \epsilon_{\mathbf{k}\sigma} & -\Delta_{\mathbf{k}\sigma} \\ -\Delta^*_{\mathbf{k}\sigma} & -\epsilon_{\mathbf{k}\sigma} \end{bmatrix}$$

and $\Delta_{\mathbf{k}\sigma} = (-1)^{\delta_{\sigma=\downarrow}} mU$. Nambu spinorial formulation is used,

$$\hat{\Psi}_{\mathbf{k}\sigma} \equiv \begin{bmatrix} \hat{c}_{\mathbf{k}\sigma} \\ \hat{c}_{\mathbf{k}+\boldsymbol{\pi}\sigma} \end{bmatrix}$$

and the free electrons energy is simply the tight binding energy

$$\epsilon_{\mathbf{k}\sigma} = -2t \left(\cos k_x + \cos k_y \right)$$

which is spin-invariant. The MFT description of the model reduces to a gas of free "γ-fermions", described by the Nambu spinor of Eq. (??),

$$\hat{\Gamma}_{\mathbf{k}\sigma} = W_{\mathbf{k}\sigma} \hat{\Psi}_{\mathbf{k}\sigma} = \begin{bmatrix} \hat{\gamma}^{(+)}_{\mathbf{k}\sigma} \\ \hat{\gamma}^{(-)}_{\mathbf{k}\sigma} \end{bmatrix}$$

where

$$W_{\mathbf{k}\sigma} = \begin{bmatrix} \cos\theta_{\mathbf{k}\sigma} & -\sin\theta_{\mathbf{k}\sigma} \\ \sin\theta_{\mathbf{k}\sigma} & \cos\theta_{\mathbf{k}\sigma} \end{bmatrix} \quad \text{and} \quad \sin 2\theta_{\mathbf{k}\sigma} \equiv \frac{\Delta_{\mathbf{k}\sigma}}{E_{\mathbf{k}\sigma}}$$

These fermions populate the two bands $\pm E_{\mathbf{k}\sigma} = \sqrt{\epsilon^2_{\mathbf{k}\sigma} + |\Delta_{\mathbf{k}\sigma}|^2}$. The entire system is mapped onto an ensemble of pseudo-spins, each subject to a pseudo-field, as in Fig. 2.1. To diagonalize the system essentially means to anti-align each pseudo-spin with the z axis. Each pseudospin component expectation value (EV) is described by a simple equation and has a precise meaning, listed in Tab. 2.1.

2.1.2 Extension to the EHM

Consider now the non-local interaction $\hat{H}_V$: since only translational invariance is partially broken in the AF phase, the only RWCs within the MFT decomposition are Hartree terms and the same-spin Fock term. The discussion is thus essentially the same as for the normal phase of Chap. 1, with the only relevant difference being the Ansatz we use in Eq. (1.1), substituted by the commensurate oscillatory Ansatz of Eq. (??). As for the normal phase, by design the MFT solution to the model for the given phase is described by a redefinition of the various physical quantities and mathematical objects,

$$\epsilon_{\mathbf{k}\sigma} \to \tilde{\epsilon}_{\mathbf{k}\sigma} \qquad E_{\mathbf{k}\sigma} \to \tilde{E}_{\mathbf{k}\sigma} \qquad \Delta_{\mathbf{k}\sigma} \to \tilde{\Delta}_{\mathbf{k}\sigma} \qquad \mu \to \tilde{\mu}$$

The band energies renormalization is simply

$$\tilde{E}_{\mathbf{k}\sigma} \equiv \sqrt{\tilde{\epsilon}^2_{\mathbf{k}\sigma} + |\tilde{\Delta}_{\mathbf{k}\sigma}|^2}$$

and the following relations hold (the *tilde* sign indicates the renormalized quantity):

$$\langle \hat{\Psi}^\dagger_{\mathbf{k}\sigma} \tau^x \hat{\Psi}_{\mathbf{k}\sigma} \rangle = -\sin(2\tilde{\theta}_{\mathbf{k}\sigma}) \sin(2\tilde{\zeta}_{\mathbf{k}\sigma}) \langle \hat{\Gamma}^\dagger_{\mathbf{k}\sigma} \tau^z \hat{\Gamma}_{\mathbf{k}\sigma} \rangle \tag{2.1}$$

$$\langle \hat{\Psi}^\dagger_{\mathbf{k}\sigma} \tau^y \hat{\Psi}_{\mathbf{k}\sigma} \rangle = -\sin(2\tilde{\theta}_{\mathbf{k}\sigma}) \cos(2\tilde{\zeta}_{\mathbf{k}\sigma}) \langle \hat{\Gamma}^\dagger_{\mathbf{k}\sigma} \tau^z \hat{\Gamma}_{\mathbf{k}\sigma} \rangle \tag{2.2}$$

$$\langle \hat{\Psi}^\dagger_{\mathbf{k}\sigma} \tau^z \hat{\Psi}_{\mathbf{k}\sigma} \rangle = \cos(2\tilde{\theta}_{\mathbf{k}\sigma}) \langle \hat{\Gamma}^\dagger_{\mathbf{k}\sigma} \tau^z \hat{\Gamma}_{\mathbf{k}\sigma} \rangle \tag{2.3}$$

while in the tilted frame, the z component of the pseudospin is just a thermal superposition of the "up" and "down" states

$$\langle \hat{\Gamma}^\dagger_{\mathbf{k}\sigma} \tau^z \hat{\Gamma}_{\mathbf{k}\sigma} \rangle = \frac{1}{2} \left[f\left(\tilde{E}_{\mathbf{k}\sigma}; \beta, \tilde{\mu}\right) - f\left(-\tilde{E}_{\mathbf{k}\sigma}; \beta, \tilde{\mu}\right) \right] \tag{2.4}$$

These equations are identical to the ones for the pure HM, Eqns. (**??**), (**??**) and (**??**). Let us define the (renormalized) thermal factors as in Eq. (**??**),

$$\tilde{\mathrm{Th}}^{(\pm)}_{\mathbf{k}\sigma}[\beta, \tilde{\mu}] \equiv f\left(\tilde{E}_{\mathbf{k}\sigma}; \beta, \tilde{\mu}\right) \pm f\left(-\tilde{E}_{\mathbf{k}\sigma}; \beta, \tilde{\mu}\right) \tag{2.5}$$

The angles satisfy:

$$\sin(2\tilde{\xi}_{\mathbf{k}\sigma}) = \frac{\mathrm{Re}\{\tilde{\Delta}_{\mathbf{k}\sigma}\}}{|\tilde{\Delta}_{\mathbf{k}\sigma}|} \quad \cos(2\tilde{\xi}_{\mathbf{k}\sigma}) = \frac{\mathrm{Im}\{\tilde{\Delta}_{\mathbf{k}\sigma}\}}{|\tilde{\Delta}_{\mathbf{k}\sigma}|} \quad \sin(2\tilde{\theta}_{\mathbf{k}\sigma}) = \frac{|\tilde{\Delta}_{\mathbf{k}\sigma}|}{\tilde{E}_{\mathbf{k}\sigma}} \quad \cos(2\tilde{\theta}_{\mathbf{k}\sigma}) = \frac{\tilde{\epsilon}_{\mathbf{k}\sigma}}{\tilde{E}_{\mathbf{k}\sigma}} \tag{2.6}$$

The physical behaviour is the same as for the pure Hubbard model. In terms of MFT discussion, the AF phase is similar enough to the normal phase of Chap. 1: the RWCs are the same of Sec. **??**, listed in Tabs. **??** and **??**. All three Hartree contractions, responsible in the normal phase of the chemical potential shift, and the s.s. Fock contraction, that originated the hopping renormalization. Let us start by the Hartree terms: due to the "staggered" nature of the commensurate AF phase, their role is here more complex. Then we will move to the Fock term, discussing the MFT approximations effect on bands. Next sections are basically an extension of the calculations of Chap. 1.

2.2 Hartree effects: chemical potential renormalization

As in Eq. (1.3), the same-spin and opposite-spin non-local Hartree terms are

$$\hat{H}_V \simeq \overbrace{-V \sum_{\langle ij \rangle} \sum_{\sigma} \left[\langle \hat{n}_{i\sigma} \rangle \hat{n}_{j\sigma} + \hat{n}_{i\sigma} \langle \hat{n}_{j\sigma} \rangle - \langle \hat{n}_{i\sigma} \rangle \langle \hat{n}_{j\sigma} \rangle \right]}^{\text{s.s.}}$$
$$\underbrace{-V \sum_{\langle ij \rangle} \sum_{\sigma} \left[\langle \hat{n}_{i\sigma} \rangle \hat{n}_{j\overline{\sigma}} + \hat{n}_{i\sigma} \langle \hat{n}_{j\overline{\sigma}} \rangle - \langle \hat{n}_{i\sigma} \rangle \langle \hat{n}_{j\overline{\sigma}} \rangle \right]}_{\text{o.s.}} + (\text{all the rest})$$

Let $i \to \mathbf{r} = (x, y)$ and $j \to \mathbf{r}' = (x', y')$. The key difference in the present derivation, with respect to the normal phase, is the single-site density Ansatz: instead of Eq. (1.1), we use Eq. (**??**), which describes an oscillatory density for each spin sector,

$$\langle \hat{n}_{\mathbf{r}\sigma} \rangle = n - (-1)^{x+y+\delta_{\sigma=\uparrow}} m$$

This Ansatz is composed of two parts: the offset n, and the oscillatory part m. The algebraic derivation relative to the n part is, of course, the same as for the normal phase. We now treat separately the "operatorial" part of the braced terms of Eq. (1.3),

$$-V \sum_{\langle \mathbf{r}\mathbf{r}' \rangle} \sum_{\sigma} \left[\langle \hat{n}_{\mathbf{r}\sigma} \rangle \hat{n}_{\mathbf{r}'\sigma} + \hat{n}_{\mathbf{r}\sigma} \langle \hat{n}_{\mathbf{r}'\sigma} \rangle \right] - V \sum_{\langle \mathbf{r}\mathbf{r}' \rangle} \sum_{\sigma} \left[\langle \hat{n}_{\mathbf{r}\sigma} \rangle \hat{n}_{\mathbf{r}'\overline{\sigma}} + \hat{n}_{\mathbf{r}\sigma} \langle \hat{n}_{\mathbf{r}'\overline{\sigma}} \rangle \right] \tag{2.7}$$

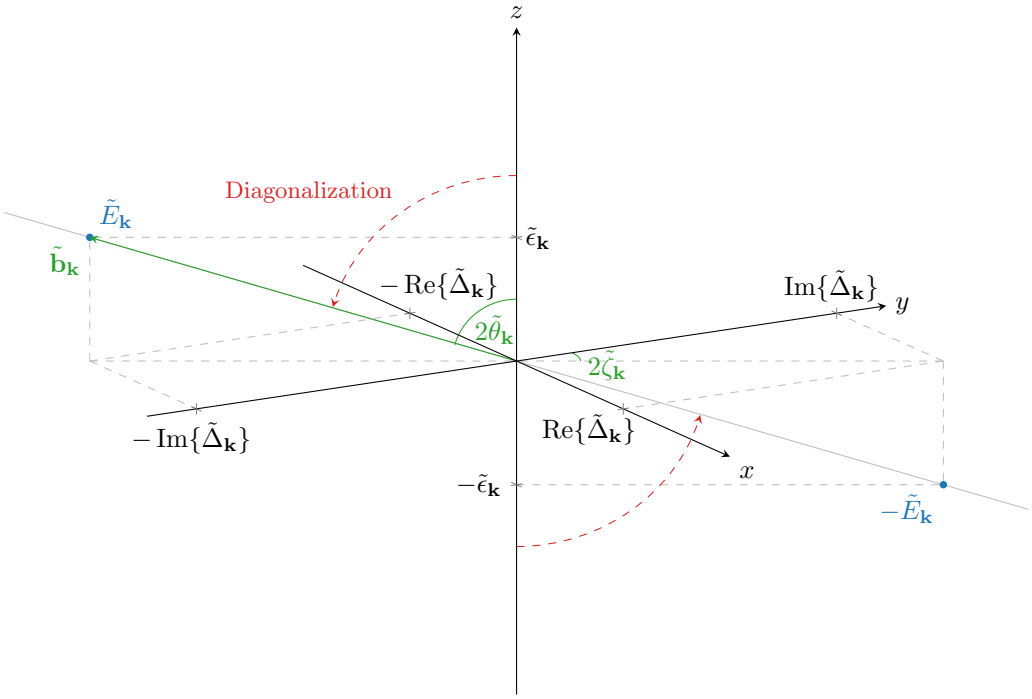


Figure 2.1 Sketch of the diagonalization of the pseudo-spin problem. The dashed line represents the diagonalization given by the z-axis alignment with the field direction. In the top-right side of the picture are listed the expectation values of the original Nambu spinor. For simplicity everywhere the index $\sigma =\uparrow$ has been omitted.

as well as the its scalar counterpart

$$V \sum_{\langle \mathbf{r}\mathbf{r}' \rangle} \sum_{\sigma} \langle \hat{n}_{\mathbf{r}\sigma} \rangle \langle \hat{n}_{\mathbf{r}'\sigma} \rangle + V \sum_{\langle \mathbf{r}\mathbf{r}' \rangle} \sum_{\sigma} \langle \hat{n}_{\mathbf{r}\sigma} \rangle \langle \hat{n}_{\mathbf{r}'\overline{\sigma}} \rangle \tag{2.8}$$

which accounts for the energy shift to be accounted when performing Wick's contractions.

2.2.1 Hartree channel in the AF phase: analytical derivation

Let us start from the operator of Expr. (2.7). By inserting the Ansatz of Eq. (**??**), we get

$$\overbrace{-nV \sum_{\langle \mathbf{r}\mathbf{r}' \rangle} \sum_{\sigma} [\hat{n}_{\mathbf{r}'\sigma} + \hat{n}_{\mathbf{r}\sigma}] + mV \sum_{\langle \mathbf{r}\mathbf{r}' \rangle} \sum_{\sigma} (-1)^{\delta_{\sigma=\uparrow}} \left[(-1)^{x+y} \hat{n}_{\mathbf{r}'\sigma} + (-1)^{x'+y'} \hat{n}_{\mathbf{r}\sigma} \right]}^{\text{s.s.}}$$
$$\underbrace{-nV \sum_{\langle \mathbf{r}\mathbf{r}' \rangle} \sum_{\sigma} [\hat{n}_{\mathbf{r}'\overline{\sigma}} + \hat{n}_{\mathbf{r}\sigma}] + mV \sum_{\langle \mathbf{r}\mathbf{r}' \rangle} \sum_{\sigma} \left[(-1)^{x+y+\delta_{\sigma=\uparrow}} \hat{n}_{\mathbf{r}'\overline{\sigma}} + (-1)^{x'+y'+\delta_{\overline{\sigma}=\uparrow}} \hat{n}_{\mathbf{r}\sigma} \right]}_{\text{o.s.}}$$

For a square lattice, if $\mathbf{r} = (x, y)$ and $\mathbf{r}' = (x', y')$ are NNs, evidently

$$(-1)^{x'+y'} = (-1)^{x+y+1}$$

Moreover,

$$(-1)^{\delta_{\overline{\sigma}=\uparrow}} = (-1)^{\delta_{\sigma=\uparrow}+1}$$

We obtain

$$\overbrace{-nV\sum_{\langle \mathbf{r}\mathbf{r}'\rangle}\sum_{\sigma}[\hat{n}_{\mathbf{r}\sigma}+\hat{n}_{\mathbf{r}'\sigma}]}^{\text{s.s. (density)}}-\overbrace{mV\sum_{\langle \mathbf{r}\mathbf{r}'\rangle}\sum_{\sigma}(-1)^{x+y+\delta_{\sigma=\uparrow}}[\hat{n}_{\mathbf{r}\sigma}-\hat{n}_{\mathbf{r}'\sigma}]}^{\text{s.s. (magnetization)}}$$
$$\underbrace{-nV\sum_{\langle \mathbf{r}\mathbf{r}'\rangle}\sum_{\sigma}[\hat{n}_{\mathbf{r}\sigma}+\hat{n}_{\mathbf{r}'\overline{\sigma}}]}_{\text{o.s. (density)}}+\underbrace{mV\sum_{\langle \mathbf{r}\mathbf{r}'\rangle}\sum_{\sigma}(-1)^{x+y+\delta_{\sigma=\uparrow}}[\hat{n}_{\mathbf{r}\sigma}+\hat{n}_{\mathbf{r}'\overline{\sigma}}]}_{\text{o.s. (magnetization)}} \tag{2.9}$$

In the above expressions the various contribution have been separated in "density" contributions and "magnetization" contributions. Let us deal with these separately.

Density terms. Consider the s.s. and o.s. (density) terms of Expr. (2.9),

$$-nV\sum_{\langle \mathbf{r}\mathbf{r}'\rangle}\sum_{\sigma}[\hat{n}_{\mathbf{r}\sigma}+\hat{n}_{\mathbf{r}'\sigma}]-nV\sum_{\langle \mathbf{r}\mathbf{r}'\rangle}\sum_{\sigma}[\hat{n}_{\mathbf{r}\sigma}+\hat{n}_{\mathbf{r}'\overline{\sigma}}] \tag{2.10}$$

The discussion is identical to the one of Sec. **??** for the normal phase, with the renormalization of μ being the same of Eq. (1.5).

$$\tilde{\mu}\equiv\mu+n(2zV-U) \tag{2.11}$$

Within this equation we are also already implicitly considering the μ shift effect due to $\hat{H}_U$.

Magnetization terms. The s.s. and o.s. (magnetization) terms of Expr. (2.9) perfectly cancel out:

$$mV\sum_{\langle \mathbf{r}\mathbf{r}'\rangle}\sum_{\sigma}(-1)^{x+y+\delta_{\sigma=\uparrow}}[(\hat{n}_{\mathbf{r}\sigma}-\hat{n}_{\mathbf{r}'\sigma})-(\hat{n}_{\mathbf{r}\sigma}+\hat{n}_{\mathbf{r}'\overline{\sigma}})]=0$$

thus the only net effect of the Hartree RWCs affects μ. This is somewhat to be expected: in the context of the HM, the gap function emerges in its purely s-wave structure from this precise computation on Hartree terms. However, the $\hat{H}_V$ non-local interaction cannot provide a pure s-wave potential component. Effects on the gap function are to be observed somewhere else.

Let us now move to the last part of the Hartree effects on the AF phase originated by $\hat{H}_V$, which is, the energy shift due to double counting terms originated by MFT decomposition of the quartic interactions $\hat{H}_U$ and $\hat{H}_V$.

2.2.2 Hartree contraction energy

Consider the scalar Expr. (2.8): it's the sum of two parts, the first coming from s.s. sector, the second from o.s. sector.

s.s. sector double counting energy. For the s.s. part, explicitating the two spin sectors, we get

$$V\sum_{\langle \mathbf{r}\mathbf{r}'\rangle}\sum_{\sigma}\langle\hat{n}_{\mathbf{r}\sigma}\rangle\langle\hat{n}_{\mathbf{r}'\sigma}\rangle=\overbrace{V\sum_{\langle \mathbf{r}\mathbf{r}'\rangle}\Big[(n+(-1)^{x+y}m)(n+(-1)^{x'+y'}m)\Big]}^{\sigma=\uparrow}$$
$$\underbrace{+V\sum_{\langle \mathbf{r}\mathbf{r}'\rangle}\Big[(n-(-1)^{x+y}m)(n-(-1)^{x'+y'}m)\Big]}_{\sigma=\downarrow}$$

Evidently the mixed terms (those of order $\mathcal{O}(n)\times\mathcal{O}(m)$) cancel out, and considering that the sign factor is necessarily alternating on neighboiring site, the remainder is just

$$V\sum_{\langle \mathbf{r}\mathbf{r}'\rangle}\sum_{\sigma}\langle\hat{n}_{i\sigma}\rangle\langle\hat{n}_{j\sigma}\rangle=V\sum_{\langle \mathbf{r}\mathbf{r}'\rangle}2(n^2-m^2)$$

o.s. sector double counting energy. Proceeding identically on the o.s. sector, we get

$$V\sum_{\langle \mathbf{r}\mathbf{r}'\rangle}\sum_{\sigma}\langle \hat{n}_{\mathbf{r}\sigma}\rangle\langle \hat{n}_{\mathbf{r}'\overline{\sigma}}\rangle = \overbrace{V\sum_{\langle \mathbf{r}\mathbf{r}'\rangle}\Big[(n+(-1)^{x+y}m)(n-(-1)^{x'+y'}m)\Big]}^{\sigma=\uparrow} + \underbrace{V\sum_{\langle \mathbf{r}\mathbf{r}'\rangle}\Big[(n-(-1)^{x+y}m)(n+(-1)^{x'+y'}m)\Big]}_{\sigma=\downarrow}$$

which gives immediately

$$V\sum_{\langle ij\rangle}\sum_{\sigma}\langle \hat{n}_{i\sigma}\rangle\langle \hat{n}_{j\overline{\sigma}}\rangle = V\sum_{\langle \mathbf{r}\mathbf{r}'\rangle}2(n^2+m^2)$$

Hence, the total "contraction energy", given by the sum of both contributions, is identical to the normal one of Eq. (**??**) indeed:

$$E^{(\mathrm{AF})}_{\mathrm{H}/V} = 2V\sum_{\langle ij\rangle}\sum_{\sigma}n^2 = 4n^2V\times\frac{z}{2}L_xL_y \tag{2.12}$$

This implies that, when comparing normal and AF phase, we can rule out this energy contribution, being equal for both free energy densities.

2.3 Fock effects: gap and bands renormalization

We explored thoroughly the Hartree contractions effects on the MFT hamiltonian. Let us now move to the effects originated by Fock contractions. We can use the derivation of Sec. 1.3 up to Eq. (1.14),

$$\begin{aligned} V\sum_{\langle ij\rangle}\sum_{\sigma}&\Big[\langle \hat{c}^\dagger_{i\sigma}\hat{c}_{j\sigma}\rangle\hat{c}^\dagger_{j\sigma}\hat{c}_{i\sigma}+\hat{c}^\dagger_{i\sigma}\hat{c}_{j\sigma}\langle \hat{c}^\dagger_{j\sigma}\hat{c}_{i\sigma}\rangle-\langle \hat{c}^\dagger_{i\sigma}\hat{c}_{j\sigma}\rangle\langle \hat{c}^\dagger_{j\sigma}\hat{c}_{i\sigma}\rangle\Big] \\ &\overset{\mathcal{F}}{=}\frac{2V}{L_xL_y}\sum_{\mathbf{K},\mathbf{k}}\sum_{\sigma}\left[\cos(2k_x)+\cos(2k_y)\right]\langle \hat{c}^\dagger_{\mathbf{K}+\mathbf{k}\sigma}\hat{c}_{\mathbf{K}+\mathbf{k}\sigma}\rangle\hat{c}^\dagger_{\mathbf{K}-\mathbf{k}\sigma}\hat{c}_{\mathbf{K}-\mathbf{k}\sigma}-E^{(\mathrm{AF})}_{\mathrm{F}/V} \end{aligned} \tag{2.13}$$

where $E^{(\mathrm{AF})}_{\mathrm{F}/V}$ is defined as in Eq. (1.13), with the expectation value being computed on the AF (thermal) ground-state. As before, we discuss the operatorial part and the scalar part separately.

2.3.1 Fock channel in the AF phase: analytical derivation

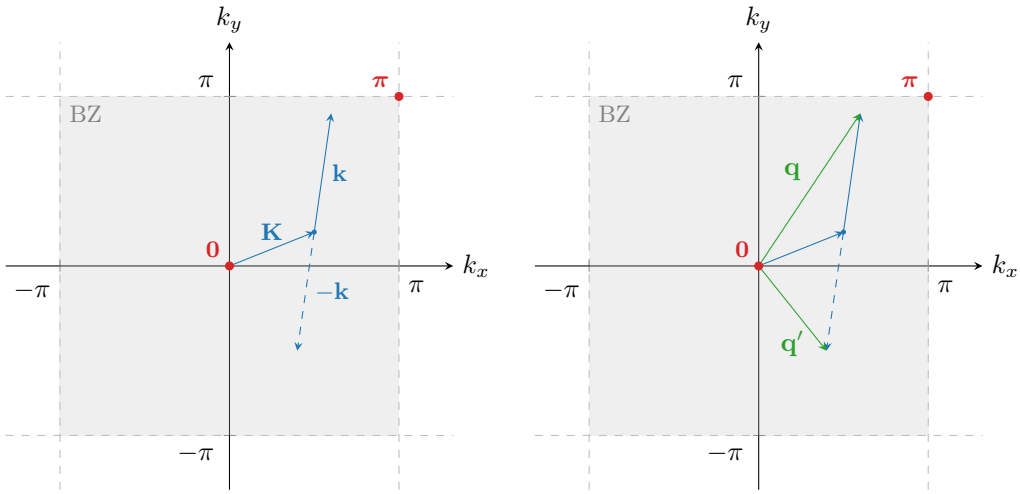
In order to proceed from Eq. (2.13), it is necessary to understand how the AF phase is realized in reciprocal space. For the conventional Hubbard model, as is exposed in App. **??**, to impose an AF Ansatz of the form

$$\langle \hat{n}_{\mathbf{r}\sigma}\rangle = n-(-1)^{x+y+\delta_{\sigma=\uparrow}}m$$

leads to an AF ground-state of free fermions at temperature β described by the Nambu spinor of Eq. (**??**). All parameters are renormalized, thus we must account for renormalized band energies $\pm\tilde{E}_{\mathbf{k}\sigma}$ as well. The ground-state is realized by simply populating the two bands $\pm\tilde{E}_{\mathbf{k}\sigma}$ as

$$\bigotimes_{\mathbf{k}\in\mathrm{MBZ}}\bigotimes_{\sigma}\left[\left(\hat{\gamma}^{(-)}_{\mathbf{k}\sigma}\right)^\dagger f(-\tilde{E}_{\mathbf{k}};\beta,\tilde{\mu})+\left(\hat{\gamma}^{(+)}_{\mathbf{k}\sigma}\right)^\dagger f(\tilde{E}_{\mathbf{k}};\beta,\tilde{\mu})\right]|\Omega\rangle$$

The $\hat{\gamma}$ operators are normalized superpositions of two $\hat{c}$ operators at points in reciprocal space separated by a $\boldsymbol{\pi}$ shift. It follows that the above state is ultimately a superposition of many-body pure states, each of which has either the $\mathbf{k}\sigma$ state occupied *or* the $\mathbf{k}+\boldsymbol{\pi}\sigma$ state for each $\mathbf{k}\in\mathrm{MBZ}$, $\sigma\in\{\uparrow,\downarrow\}$. It follows that, when computing generically $\langle \hat{c}^\dagger_{\mathbf{k}_1\sigma}\hat{c}_{\mathbf{k}_2\sigma}\rangle$, such expectation value can be



a | Generic vectors. b | Change of variables.

Figure 2.2 | Representation of the vectors involved in the diagonal terms of Eq. (**??**). In Fig. 2.2a generic vectors are considered, cycling over all values of $\mathbf{K}, \mathbf{k} \in \mathrm{BZ}$. In Fig. 2.2b is depicted the variables change to the new vectors $\mathbf{q}, \mathbf{q}'$.

non-zero if and only if $\mathbf{k}_1 = \mathbf{k}_2 + n\boldsymbol{\pi}$, being $n \in \mathbb{Z}$. Going back to Eq. (**??**), this implies only two contributions are non-zero:

$$\mathbf{k} = -\mathbf{k}' \quad \text{or} \quad \mathbf{k} + \boldsymbol{\pi} = -\mathbf{k}'$$

The first case gives us the normal fermionic EV, $\langle \hat{c}^\dagger_{\mathbf{k}\sigma} \hat{c}_{\mathbf{k}\sigma} \rangle$, the second the staggered AF order parameter, $\langle \hat{c}^\dagger_{\mathbf{k}\sigma} \hat{c}_{\mathbf{k}+\boldsymbol{\pi}\sigma} \rangle$[1]. Then Eq. (2.13) is reduced to:

$$(\cdots) \overset{\mathcal{F}}{=} \frac{2V}{L_x L_y} \sum_{\mathbf{K},\mathbf{k}} \sum_{\sigma} \left[\cos(2k_x) + \cos(2k_y)\right] \left[\underbrace{\langle \hat{c}^\dagger_{\mathbf{K}+\mathbf{k}\sigma} \hat{c}_{\mathbf{K}+\mathbf{k}\sigma} \rangle \hat{c}^\dagger_{\mathbf{K}-\mathbf{k}\sigma} \hat{c}_{\mathbf{K}-\mathbf{k}\sigma}}_{\text{Diagonal terms}} - \underbrace{\langle \hat{c}^\dagger_{\mathbf{K}+\mathbf{k}\sigma} \hat{c}_{\mathbf{K}+\mathbf{k}+\boldsymbol{\pi}\sigma} \rangle \hat{c}^\dagger_{\mathbf{K}-\mathbf{k}\sigma} \hat{c}_{\mathbf{K}-\mathbf{k}-\boldsymbol{\pi}\sigma}}_{\text{Off-diagonal terms}} \right] \tag{2.14}$$

Now, the above equation presents *diagonal* and *off-diagonal* terms.

Diagonal terms. The diagonal terms of Eq. (2.14) are simple density interactions with the mean density field. Consider Fig. 2.2a: density at vector $\mathbf{q} \equiv \mathbf{K} + \mathbf{k}$ interacts with the mean density at vector $\mathbf{q}' \equiv \mathbf{K} - \mathbf{k}$. These variables are depicted in Fig. 2.2b. Apply in the diagonal part of Eq. (2.14) this variables change (which is the same of Eq. (1.15))

$$\begin{aligned} &\frac{2V}{L_x L_y} \sum_{\mathbf{q},\mathbf{q}'} \sum_{\sigma} [\cos(2k_x) + \cos(2k_y)] \langle \hat{c}^\dagger_{\mathbf{K}+\mathbf{k}\sigma} \hat{c}_{\mathbf{K}+\mathbf{k}\sigma} \rangle \hat{c}^\dagger_{\mathbf{K}-\mathbf{k}\sigma} \hat{c}_{\mathbf{K}-\mathbf{k}\sigma} \\ &\qquad = \frac{2V}{L_x L_y} \sum_{\mathbf{q},\mathbf{q}'} \sum_{\sigma} \left[\cos(\delta q_x) + \cos(\delta q_y)\right] \langle \hat{c}^\dagger_{\mathbf{q}\sigma} \hat{c}_{\mathbf{q}\sigma} \rangle \hat{c}^\dagger_{\mathbf{q}'\sigma} \hat{c}_{\mathbf{q}'\sigma} \\ &\qquad \overset{*}{=} \sum_{\gamma,\sigma} \left[\frac{1}{L_x L_y} \sum_{\mathbf{q}} \varphi^{(\gamma)}_{\mathbf{q}} \langle \hat{c}^\dagger_{\mathbf{q}\sigma} \hat{c}_{\mathbf{q}\sigma} \rangle \right] \times V \sum_{\mathbf{q}'} \varphi^{(\gamma)*}_{\mathbf{q}'} \hat{c}^\dagger_{\mathbf{q}'\sigma} \hat{c}_{\mathbf{q}'\sigma} \end{aligned} \tag{2.15}$$

where the sign $\overset{*}{=}$ indicates that we have substituted the decomposition of Eq. (**??**). Define as in Eq. (1.17) the EV

$$u_{\mathbf{q}\sigma} \equiv \langle \hat{c}^\dagger_{\mathbf{q}\sigma} \hat{c}_{\mathbf{q}\sigma} \rangle \tag{2.16}$$

[1] All of this is equivalent to saying that we are explicitly breaking symmetry using the order parameter $\langle \hat{c}^\dagger_{\mathbf{k}\sigma} \hat{c}_{\mathbf{k}+\boldsymbol{\pi}\sigma} \rangle$, and *thus* the state is realized as described in text.

The expression in square brackets of Eq. (2.15) is just its γ component, and is expanded in

$$\begin{aligned}
u_\sigma^{(\gamma)} &= \frac{1}{L_x L_y} \sum_{\mathbf{q}\in\mathrm{BZ}} \varphi_{\mathbf{q}}^{(\gamma)} \langle \hat{c}^\dagger_{\mathbf{q}\sigma} \hat{c}_{\mathbf{q}\sigma} \rangle \\
&= \frac{1}{L_x L_y} \sum_{\mathbf{q}\in\mathrm{MBZ}} \left[\varphi_{\mathbf{q}}^{(\gamma)} \langle \hat{c}^\dagger_{\mathbf{q}\sigma} \hat{c}_{\mathbf{q}\sigma} \rangle + \varphi_{\mathbf{q}+\boldsymbol{\pi}}^{(\gamma)} \langle \hat{c}^\dagger_{\mathbf{q}+\boldsymbol{\pi}\sigma} \hat{c}_{\mathbf{q}+\boldsymbol{\pi}\sigma} \rangle \right] \\
&= \frac{1}{L_x L_y} \sum_{\mathbf{q}\in\mathrm{MBZ}} \varphi_{\mathbf{q}}^{(\gamma)} \left[\langle \hat{c}^\dagger_{\mathbf{q}\sigma} \hat{c}_{\mathbf{q}\sigma} \rangle - \langle \hat{c}^\dagger_{\mathbf{q}+\boldsymbol{\pi}\sigma} \hat{c}_{\mathbf{q}+\boldsymbol{\pi}\sigma} \rangle \right] \\
&= \frac{1}{L_x L_y} \sum_{\mathbf{q}\in\mathrm{MBZ}} \varphi_{\mathbf{q}}^{(\gamma)} \langle \hat{\Psi}^\dagger_{\mathbf{q}\sigma} \tau^z \hat{\Psi}_{\mathbf{q}\sigma} \rangle \\
&= \frac{1}{2L_x L_y} \sum_{\mathbf{q}\in\mathrm{MBZ}} \varphi_{\mathbf{q}}^{(\gamma)} \frac{\tilde{\epsilon}_{\mathbf{q}\sigma}}{\tilde{E}_{\mathbf{q}\sigma}} \tilde{\mathrm{Th}}^{(-)}_{\mathbf{q}\sigma}[\beta, \tilde{\mu}]
\end{aligned} \tag{2.17}$$

where, in the last passage, Eqns. (2.3), (2.4) and (2.5) have been used. Now, the AF phase does not break inversion symmetry: this implies that, whatever the shape of the renormalized bands, it must be

$$\tilde{\epsilon}_{\mathbf{q}\sigma} = \tilde{\epsilon}_{-\mathbf{q}\sigma} \quad \Longrightarrow \quad u_\sigma^{(p_\ell)} = 0 \quad \text{with} \quad \ell \in \{x, y\}$$

The renormalized bands need to be symmetric functions of the moment. Hence for $\gamma \in \{p_x, p_y\}$ the above sum vanishes. Thus, Eq. (**??**) becomes

$$\begin{aligned}
&\frac{2V}{L_x L_y} \sum_{\mathbf{q},\mathbf{q}'} \sum_\sigma \left[\cos(2k_x) + \cos(2k_y)\right] \langle \hat{c}^\dagger_{\mathbf{K}+\mathbf{k}\sigma} \hat{c}_{\mathbf{K}+\mathbf{k}\sigma} \rangle \hat{c}^\dagger_{\mathbf{K}-\mathbf{k}\sigma} \hat{c}_{\mathbf{K}-\mathbf{k}\sigma} \\
&\qquad = V \sum_{\mathbf{q},\sigma} \left[u_\sigma^{(s^*)} (\cos q_x + \cos q_y) + u_\sigma^{(d)} (\cos q_x - \cos q_y) \right] \hat{c}^\dagger_{\mathbf{q}\sigma} \hat{c}_{\mathbf{q}\sigma}
\end{aligned}$$

These two factors, together, define the full bands renormalization

$$\tilde{\epsilon}_{\mathbf{k}\sigma} \equiv \epsilon_{\mathbf{k}\sigma} + u_\sigma^{(s^*)} V (\cos q_x + \cos q_y) + u_\sigma^{(d)} V (\cos q_x - \cos q_y)$$

The s^*-wave component is particularly handy, because it can interpreted as a rigid hopping shift,

$$t \to \tilde{t}_\sigma = t - \frac{1}{2} u_\sigma^{(s^*)} V$$

As for the normal phase, this RMP is controlled linearly by V and can lead to a relevant reduction of the bare bands diffusive character.

Through this decomposition we have identified two HFPs and their self-consistency equations,

$$u_\sigma^{(s^*)} = \frac{1}{2L_x L_y} \sum_{\mathbf{q}\in\mathrm{MBZ}} (\cos q_x + \cos q_y) \frac{\tilde{\epsilon}_{\mathbf{q}\sigma}}{\tilde{E}_{\mathbf{q}\sigma}} \tilde{\mathrm{Th}}^{(-)}_{\mathbf{q}\sigma}[\beta, \tilde{\mu}] \tag{2.18}$$

$$u_\sigma^{(d)} = \frac{1}{2L_x L_y} \sum_{\mathbf{q}\in\mathrm{MBZ}} (\cos q_x - \cos q_y) \frac{\tilde{\epsilon}_{\mathbf{q}\sigma}}{\tilde{E}_{\mathbf{q}\sigma}} \tilde{\mathrm{Th}}^{(-)}_{\mathbf{q}\sigma}[\beta, \tilde{\mu}] \tag{2.19}$$

The general HF scheme, described in Sec. [Missing], is iterative and aims to use these equations iteratively to find a stable solution for these parameters.

Off-diagonal terms. Consider the off-diagonal terms of Eq. (2.14). These contribute instead to the full gap, being out of diagonal in the 2×2 hamiltonian matrix. In terms of the pseudospin picture, they account for a transverse field. Define $\mathbf{q}$, $\mathbf{q}'$ as in Fig. 2.2b, and rewrite as for Eq. (2.15)

$$\begin{aligned}
&-\frac{2V}{L_x L_y} \sum_{\mathbf{q},\mathbf{q}'} \sum_\sigma [\cos(2k_x) + \cos(2k_y)] \langle \hat{c}^\dagger_{\mathbf{K}+\mathbf{k}+\boldsymbol{\pi}\sigma} \hat{c}_{\mathbf{K}+\mathbf{k}\sigma} \rangle \hat{c}^\dagger_{\mathbf{K}-\mathbf{k}\sigma} \hat{c}_{\mathbf{K}-\mathbf{k}-\boldsymbol{\pi}\sigma} \\
&\qquad = -\sum_{\gamma,\sigma} \left[\frac{1}{L_x L_y} \sum_{\mathbf{q}} \varphi_{\mathbf{q}}^{(\gamma)} \langle \hat{c}^\dagger_{\mathbf{q}\sigma} \hat{c}_{\mathbf{q}+\boldsymbol{\pi}\sigma} \rangle \right] \times V \sum_{\mathbf{q}'} \varphi_{\mathbf{q}'}^{(\gamma)*} \hat{c}^\dagger_{\mathbf{q}'\sigma} \hat{c}_{\mathbf{q}'+\boldsymbol{\pi}\sigma}
\end{aligned} \tag{2.20}$$

Proceed as done previously: define the expectation value

$$v_{\mathbf{q}\sigma} \equiv i\langle \hat{c}^\dagger_{\mathbf{q}\sigma}\hat{c}_{\mathbf{q}+\boldsymbol{\pi}\sigma}\rangle \tag{2.21}$$

(which is just the AF order parameter) and its γ-wave component, with $\gamma \in \{s^*, d, p_x, p_y\}$:

$$\begin{aligned}
v^{(\gamma)}_\sigma &= \frac{i}{L_xL_y}\sum_{\mathbf{q}\in\mathrm{BZ}} \varphi^{(\gamma)}_{\mathbf{q}}\langle \hat{c}^\dagger_{\mathbf{q}\sigma}\hat{c}_{\mathbf{q}+\boldsymbol{\pi}\sigma}\rangle \\
&= \frac{i}{L_xL_y}\sum_{\mathbf{q}\in\mathrm{MBZ}} \left[\varphi^{(\gamma)}_{\mathbf{q}}\langle \hat{c}^\dagger_{\mathbf{q}\sigma}\hat{c}_{\mathbf{q}+\boldsymbol{\pi}\sigma}\rangle + \varphi^{(\gamma)}_{\mathbf{q}+\boldsymbol{\pi}}\langle \hat{c}^\dagger_{\mathbf{q}+\boldsymbol{\pi}\sigma}\hat{c}_{\mathbf{q}\sigma}\rangle\right] \\
&= \frac{i}{L_xL_y}\sum_{\mathbf{q}\in\mathrm{MBZ}} \varphi^{(\gamma)}_{\mathbf{q}}\left[\langle \hat{c}^\dagger_{\mathbf{q}\sigma}\hat{c}_{\mathbf{q}+\boldsymbol{\pi}\sigma}\rangle - \langle \hat{c}^\dagger_{\mathbf{q}+\boldsymbol{\pi}\sigma}\hat{c}_{\mathbf{q}\sigma}\rangle\right] \\
&= -\frac{1}{L_xL_y}\sum_{\mathbf{q}\in\mathrm{MBZ}} \varphi^{(\gamma)}_{\mathbf{q}}\langle \hat{\Psi}^\dagger_{\mathbf{q}\sigma}\tau^y\hat{\Psi}_{\mathbf{q}\sigma}\rangle \\
&= \frac{1}{2L_xL_y}\sum_{\mathbf{q}\in\mathrm{MBZ}} \varphi^{(\gamma)}_{\mathbf{q}}\frac{\mathrm{Im}\{\tilde{\Delta}_{\mathbf{q}\sigma}\}}{\tilde{E}_{\mathbf{q}\sigma}}\tilde{\mathrm{Th}}^{(-)}_{\mathbf{q}\sigma}[\beta,\tilde{\mu}]
\end{aligned} \tag{2.22}$$

From the definitions of the structure factors of Tab. **??**, it's immediate to see that:

$$v^{(s^*)}_\sigma, v^{(d)}_\sigma \in \mathbb{R} \qquad v^{(p_x)}_\sigma, v^{(p_y)}_\sigma \in i\mathbb{R}$$

Getting back to Eq. (2.20) and inserting these components, we get

$$\begin{aligned}
-\frac{2V}{L_xL_y}\sum_{\mathbf{q},\mathbf{q}'}\sum_\sigma[\cos(2k_x)+\cos(2k_y)]\langle \hat{c}^\dagger_{\mathbf{K}+\mathbf{k}+\boldsymbol{\pi}\sigma}\hat{c}_{\mathbf{K}+\mathbf{k}\sigma}\rangle \hat{c}^\dagger_{\mathbf{K}-\mathbf{k}\sigma}\hat{c}_{\mathbf{K}-\mathbf{k}-\boldsymbol{\pi}\sigma} \\
= iV\sum_{\gamma,\sigma} v^{(\gamma)}_\sigma \sum_{\mathbf{q}\in\mathrm{BZ}} \varphi^{(\gamma)*}_{\mathbf{q}}\hat{c}^\dagger_{\mathbf{q}\sigma}\hat{c}_{\mathbf{q}+\boldsymbol{\pi}\sigma}
\end{aligned} \tag{2.23}$$

We can expand on the MBZ as

$$\begin{aligned}
-\frac{2V}{L_xL_y}\sum_{\mathbf{q},\mathbf{q}'}\sum_\sigma[\cos(2k_x)+\cos(2k_y)]\langle \hat{c}^\dagger_{\mathbf{K}+\mathbf{k}+\boldsymbol{\pi}\sigma}\hat{c}_{\mathbf{K}+\mathbf{k}\sigma}\rangle \hat{c}^\dagger_{\mathbf{K}-\mathbf{k}\sigma}\hat{c}_{\mathbf{K}-\mathbf{k}-\boldsymbol{\pi}\sigma} \\
= iV\sum_{\gamma,\sigma}\sum_{\mathbf{q}\in\mathrm{MBZ}}\left[v^{(\gamma)}_\sigma\varphi^{(\gamma)*}_{\mathbf{q}}\hat{c}^\dagger_{\mathbf{q}\sigma}\hat{c}_{\mathbf{q}+\boldsymbol{\pi}\sigma} + v^{(\gamma)}_\sigma\varphi^{(\gamma)*}_{\mathbf{q}+\boldsymbol{\pi}}\hat{c}^\dagger_{\mathbf{q}+\boldsymbol{\pi}\sigma}\hat{c}_{\mathbf{q}\sigma}\right]
\end{aligned} \tag{2.24}$$

Recalling Tab. **??**, for the symmetries we are considering

$$iv^{(\gamma)}_\sigma\varphi^{(\gamma)*}_{\mathbf{q}} \in i\mathbb{R}$$

Moreover, since $\varphi^{(\gamma)}_{\mathbf{q}} = -\varphi^{(\gamma)}_{\mathbf{q}+\boldsymbol{\pi}}$, we have

$$iv^{(\gamma)}_\sigma\varphi^{(\gamma)*}_{\mathbf{q}+\boldsymbol{\pi}} = \left(iv^{(\gamma)}_\sigma\varphi^{(\gamma)*}_{\mathbf{q}}\right)^*$$

Therefore

$$\begin{aligned}
-\frac{2V}{L_xL_y}\sum_{\mathbf{q},\mathbf{q}'}\sum_\sigma[\cos(2k_x)+\cos(2k_y)]\langle \hat{c}^\dagger_{\mathbf{K}+\mathbf{k}+\boldsymbol{\pi}\sigma}\hat{c}_{\mathbf{K}+\mathbf{k}\sigma}\rangle \hat{c}^\dagger_{\mathbf{K}-\mathbf{k}\sigma}\hat{c}_{\mathbf{K}-\mathbf{k}-\boldsymbol{\pi}\sigma} \\
= V\sum_{\gamma,\sigma}\sum_{\mathbf{q}\in\mathrm{MBZ}}\left[iv^{(\gamma)}_\sigma\varphi^{(\gamma)*}_{\mathbf{q}}\hat{c}^\dagger_{\mathbf{q}\sigma}\hat{c}_{\mathbf{q}+\boldsymbol{\pi}\sigma} + \text{h.c.}\right]
\end{aligned} \tag{2.25}$$

Four more real HFPs are then identified, with their self-consistency equations,

$$v_\sigma^{(s^*)} = \frac{1}{2L_xL_y}\sum_{\mathbf{q}\in\mathrm{MBZ}}(\cos q_x+\cos q_y)\frac{\mathrm{Im}\{\tilde{\Delta}_{\mathbf{q}\sigma}\}}{\tilde{E}_{\mathbf{q}\sigma}}\tilde{\mathrm{Th}}^{(-)}_{\mathbf{q}\sigma}[\beta,\tilde{\mu}] \tag{2.26}$$

$$v_\sigma^{(d)} = \frac{1}{2L_xL_y}\sum_{\mathbf{q}\in\mathrm{MBZ}}(\cos q_x-\cos q_y)\frac{\mathrm{Im}\{\tilde{\Delta}_{\mathbf{q}\sigma}\}}{\tilde{E}_{\mathbf{q}\sigma}}\tilde{\mathrm{Th}}^{(-)}_{\mathbf{q}\sigma}[\beta,\tilde{\mu}] \tag{2.27}$$

$$-iv_\sigma^{(p_x)} = \frac{1}{\sqrt{2}L_xL_y}\sum_{\mathbf{q}\in\mathrm{MBZ}}\sin q_x\frac{\mathrm{Im}\{\tilde{\Delta}_{\mathbf{q}\sigma}\}}{\tilde{E}_{\mathbf{q}\sigma}}\tilde{\mathrm{Th}}^{(-)}_{\mathbf{q}\sigma}[\beta,\tilde{\mu}] \tag{2.28}$$

$$-iv_\sigma^{(p_y)} = \frac{1}{\sqrt{2}L_xL_y}\sum_{\mathbf{q}\in\mathrm{MBZ}}\sin q_y\frac{\mathrm{Im}\{\tilde{\Delta}_{\mathbf{q}\sigma}\}}{\tilde{E}_{\mathbf{q}\sigma}}\tilde{\mathrm{Th}}^{(-)}_{\mathbf{q}\sigma}[\beta,\tilde{\mu}] \tag{2.29}$$

As we will discuss later in Sec. 2.4.1, not all these HFPs can coexist simultaneously. For now, let us move to the free energy density computation.

2.3.2 Fock contraction energy

Let us now get back to $E^{(\mathrm{AF})}_{\mathrm{F}/V}$, the double counting energy of Eq. (2.13). As said before, its definition is the same as for the normal phase, Eq. (1.13), with the averaging operation being performed on the AF ground state:

$$E^{(\mathrm{AF})}_{\mathrm{F}/V} \equiv \frac{V}{L_xL_y}\sum_{\mathbf{K},\mathbf{k},\mathbf{k}'}\sum_\sigma[\cos(\delta k_x)+\cos(\delta k_y)]\langle\hat{c}^\dagger_{\mathbf{K}+\mathbf{k}\sigma}\hat{c}_{\mathbf{K}-\mathbf{k}'\sigma}\rangle\langle\hat{c}^\dagger_{\mathbf{K}-\mathbf{k}\sigma}\hat{c}_{\mathbf{K}+\mathbf{k}'\sigma}\rangle \tag{2.30}$$

Proceeding exactly as in Eq. (2.14) and making the change of variables of Fig. 2.2, we get

$$E^{(\mathrm{AF})}_{\mathrm{F}/V} = \frac{V}{L_xL_y}\sum_{\mathbf{q},\mathbf{q}'}\sum_\sigma[\cos(\delta q_x)+\cos(\delta q_y)]\Big[\underbrace{\langle\hat{c}^\dagger_{\mathbf{q}\sigma}\hat{c}_{\mathbf{q}\sigma}\rangle\langle\hat{c}^\dagger_{\mathbf{q}'\sigma}\hat{c}_{\mathbf{q}'\sigma}\rangle}_{\text{Diagonal terms}}-\underbrace{\langle\hat{c}^\dagger_{\mathbf{q}\sigma}\hat{c}_{\mathbf{q}+\boldsymbol{\pi}\sigma}\rangle\langle\hat{c}^\dagger_{\mathbf{q}'\sigma}\hat{c}_{\mathbf{q}'+\boldsymbol{\pi}\sigma}\rangle}_{\text{Off-diagonal terms}}\Big]$$

Apply the decomposition of Eq. (**??**) and use Eqns. (2.16), (2.21),

$$E^{(\mathrm{AF})}_{\mathrm{F}/V} = \frac{V}{2L_xL_y}\sum_{\gamma,\sigma}\sum_{\mathbf{q},\mathbf{q}'}\varphi^{(\gamma)}_{\mathbf{q}}\varphi^{(\gamma)*}_{\mathbf{q}'}[u_{\mathbf{q}\sigma}u_{\mathbf{q}'\sigma}+v_{\mathbf{q}\sigma}v_{\mathbf{q}'\sigma}] \tag{2.31}$$

Now: it holds

$$u_{\mathbf{q}\sigma} = \langle\hat{c}^\dagger_{\mathbf{q}\sigma}\hat{c}_{\mathbf{q}\sigma}\rangle \quad\Longrightarrow\quad u_{\mathbf{q}\sigma} = u^*_{\mathbf{q}\sigma}$$

and then

$$\begin{aligned}\frac{1}{(L_xL_y)^2}\sum_{\mathbf{q},\mathbf{q}'}\varphi^{(\gamma)}_{\mathbf{q}}\varphi^{(\gamma)*}_{\mathbf{q}'}u_{\mathbf{q}\sigma}u_{\mathbf{q}'\sigma} &= \left[\frac{1}{L_xL_y}\sum_{\mathbf{q}\in\mathrm{BZ}}\varphi^{(\gamma)}_{\mathbf{q}}u_{\mathbf{q}\sigma}\right]\left[\frac{1}{L_xL_y}\sum_{\mathbf{q}'\in\mathrm{BZ}}\varphi^{(\gamma)*}_{\mathbf{q}'}u^*_{\mathbf{q}'\sigma}\right]\\ &= \left|u^{(\gamma)}_\sigma\right|^2\end{aligned}$$

At the same time,

$$v_{\mathbf{q}\sigma} = i\langle\hat{c}^\dagger_{\mathbf{q}\sigma}\hat{c}_{\mathbf{q}+\boldsymbol{\pi}\sigma}\rangle \quad\Longrightarrow\quad v_{\mathbf{q}+\boldsymbol{\pi}\sigma} = -v^*_{\mathbf{q}\sigma}$$

Since the form factors satisfy $\varphi^{(\gamma)}_{\mathbf{q}+\boldsymbol{\pi}} = -\varphi^{(\gamma)}_{\mathbf{q}}$, we get:

$$\begin{aligned}\frac{1}{(L_xL_y)^2}\sum_{\mathbf{q},\mathbf{q}'}\varphi^{(\gamma)}_{\mathbf{q}}\varphi^{(\gamma)*}_{\mathbf{q}'}v_{\mathbf{q}\sigma}v_{\mathbf{q}'\sigma} &= \left[\frac{1}{L_xL_y}\sum_{\mathbf{q}\in\mathrm{BZ}}\varphi^{(\gamma)}_{\mathbf{q}}v_{\mathbf{q}\sigma}\right]\left[\frac{1}{L_xL_y}\sum_{\mathbf{q}'\in\mathrm{BZ}}\varphi^{(\gamma)*}_{\mathbf{q}'}v_{\mathbf{q}'\sigma}\right]\\ &= v^{(\gamma)}_\sigma\left[\frac{1}{L_xL_y}\sum_{\mathbf{k}\in\mathrm{BZ}}\varphi^{(\gamma)*}_{\mathbf{k}+\boldsymbol{\pi}}v_{\mathbf{k}+\boldsymbol{\pi}\sigma}\right]\\ &= v^{(\gamma)}_\sigma\left[\frac{1}{L_xL_y}\sum_{\mathbf{k}\in\mathrm{BZ}}\varphi^{(\gamma)*}_{\mathbf{k}}v^*_{\mathbf{k}\sigma}\right]\\ &= v^{(\gamma)}_\sigma\times v^{(\gamma)*}_\sigma\end{aligned}$$

where in the second passage we employed the change of variables $\mathbf{q}' = \mathbf{k}+\boldsymbol{\pi}$ and used BZ periodicity to map the original sum onto a new sum for $\mathbf{k} \in \mathrm{BZ}$. Inserting the above results in Eq. (2.31), this last becomes:

$$E_{\mathrm{F}/V}^{(\mathrm{AF})} = L_x L_y \times \frac{V}{2} \sum_{\gamma,\sigma} \left[\left| u_\sigma^{(\gamma)} \right|^2 + \left| v_\sigma^{(\gamma)} \right|^2 \right] \tag{2.32}$$

Interestingly enough, the new HFPs amplitudes contribute to lowering the contraction energy.

2.4 Full MFT hamiltonian and gap function

Finally, it is time to collect everything in a single hamiltonian operator. The MFT model is approximated by:

$$\begin{aligned}\hat{H} \simeq \sum_\sigma \sum_{\mathbf{q}\in\mathrm{MBZ}} \tilde{\epsilon}_{\mathbf{k}\sigma} \left[\hat{c}^\dagger_{\mathbf{k}\sigma} \hat{c}_{\mathbf{k}\sigma} - \hat{c}^\dagger_{\mathbf{k}+\boldsymbol{\pi}\sigma} \hat{c}_{\mathbf{k}+\boldsymbol{\pi}\sigma} \right] \\ - \sum_\sigma \sum_{\mathbf{q}\in\mathrm{MBZ}} \left[\tilde{\Delta}_{\mathbf{k}\sigma} \hat{c}^\dagger_{\mathbf{k}\sigma} \hat{c}_{\mathbf{k}+\boldsymbol{\pi}\sigma} + \tilde{\Delta}^*_{\mathbf{k}\sigma} \hat{c}^\dagger_{\mathbf{k}+\boldsymbol{\pi}\sigma} \hat{c}_{\mathbf{k}\sigma} \right] - \left[E_{\mathrm{H}/U}^{(\mathrm{AF})} + E_{\mathrm{H}/V}^{(\mathrm{AF})} + E_{\mathrm{F}/V}^{(\mathrm{AF})} \right]\end{aligned}$$

where the renormalized bands are given by:

$$\tilde{\epsilon}_{\mathbf{k}\sigma} = \epsilon_{\mathbf{k}\sigma} + V \sum_{\gamma\in\{s^*,d\}} u_\sigma^{(\gamma)} \varphi_{\mathbf{k}}^{(\gamma)*}$$

and the full gap function by

$$\tilde{\Delta}_{\mathbf{k}\sigma} = \Delta_{\mathbf{k}\sigma} + iV \sum_{\gamma,\sigma} v_\sigma^{(\gamma)} \varphi_{\mathbf{k}}^{(\gamma)*}$$

The gap correction is purely imaginary, compared to the HM gap $\Delta_{\mathbf{k}\sigma} = (-1)^{\delta_{\sigma=\downarrow}} mU$. This has two obvious consequences: being no other contribution to the x pseudospin projections other than the pure HM ones, spin antisymmetry is preserved within such direction

$$\langle \hat{s}^x_{\mathbf{k}\sigma} \rangle = - \langle \hat{s}^x_{\mathbf{k}\overline{\sigma}} \rangle$$

which is relevant in the computation of the physical magnetization,

$$\begin{aligned} m &= \frac{1}{2L_xL_y} \sum_{\mathbf{r}} (-1)^{x+y} \langle \hat{n}_{\mathbf{r}\uparrow} - \hat{n}_{\mathbf{r}\downarrow} \rangle \\ &= \frac{1}{2L_xL_y} \sum_{\mathbf{k}\in\mathrm{BZ}} \langle \hat{c}^\dagger_{\mathbf{k}\uparrow} \hat{c}_{\mathbf{k}+\boldsymbol{\pi}\uparrow} - \hat{c}^\dagger_{\mathbf{k}\downarrow} \hat{c}_{\mathbf{k}+\boldsymbol{\pi}\downarrow} \rangle \\ &= \frac{1}{L_xL_y} \sum_{\mathbf{k}\in\mathrm{MBZ}} \langle \hat{s}^x_{\mathbf{k}\uparrow} \rangle \end{aligned}$$

where we procedeed as in Sec. **??**. Moreover, the net effect $\hat{H}_V$ is a "gap complexification". In other words,

$$\left| \mathrm{Im}\{\Delta_{\mathbf{k}\sigma}\} \right| \geq 0 \quad \Longrightarrow \quad \left| \langle s^y_{\mathbf{k}\sigma} \rangle \right| \geq 0$$

while for the HM the y pseudospin component was identically null.

2.4.1 AF phase flavours and spin degeneracy

The AF phase we are studying cannot break inversion symmetry, a core property of the system which is respected from all our approximations. This translates in the simple requirement that:

$$\tilde{E}_{\mathbf{k}\sigma} \stackrel{!}{=} \tilde{E}_{-\mathbf{k}\sigma}$$

We know that:

$$\tilde{E}_{\mathbf{k}\sigma} = \sqrt{\tilde{\epsilon}^2_{\mathbf{k}\sigma} + \mathrm{Re}^2\{\tilde{\Delta}_{\mathbf{k}\sigma}\} + \mathrm{Im}^2\{\tilde{\Delta}_{\mathbf{k}\sigma}\}}$$

Phase	HFPs
sAF	$\{m, u^{(s^*)}, u^{(d)}, v^{(s^*)}, v^{(d)}\}$
aAF	$\{m, u^{(s^*)}, u^{(d)}, -iv^{(p_x)}, -iv^{(p_y)}\}$

Table 2.2 Set of HFP for the two possible realization of the AF phase, respectively a symmetric and an antisymmetric imaginary part of the gap.

The renormalized bare bands as well as the real part of the gap satisfy the inversion symmetry requirement,

$$\tilde{\epsilon}_{\mathbf{k}\sigma} = \tilde{\epsilon}_{-\mathbf{k}\sigma} \qquad \mathrm{Re}\{\tilde{\Delta}_{\mathbf{k}\sigma}\} = \mathrm{Re}\{\tilde{\Delta}_{-\mathbf{k}\sigma}\}$$

hence inversion symmetry is respected only if

$$\mathrm{Im}\{\tilde{\Delta}_{\mathbf{k}\sigma}\} = \pm\,\mathrm{Im}\{\tilde{\Delta}_{-\mathbf{k}\sigma}\}$$

which is: either the imaginary part of the gap is inversion symmetric, or is antisymmetric. Based on this trivial consideration, we are able two discard two of the four self-consistency equations for the v-related HFPs: s^* and d-wave symmetries HFPs – Eqns. (2.26) and (2.27), respectively – are nonzero if $\mathrm{Im}\{\tilde{\Delta}_{\mathbf{k}\sigma}\}$ is momentum-even; p_x and p_y-wave symmetries HFPs – Eqns. (2.28) and (2.29), respectively – are nonzero if $\mathrm{Im}\{\tilde{\Delta}_{\mathbf{k}\sigma}\}$ is momentum-odd. The two option cannot coexist, and separate simulations can be implemented in the two symmetry sectors. We name the two possible symmetries "sAF" (symmetric antiferromagnet) and "aAF" (antisymmetric antiferromagnet). The set of HFPs then restricts in these AF structures, and is reported for both cases in Tab. 2.2.

Moreover, from now on we will omit the σ index, working just in the $\uparrow$ sector. As said in Sec. **??**, the AF phase does not break $\mathbb{Z}_2$ spin inversion symmetry, being the SDW ordering generated by a staggered but symmetric spin density distribution among sites. The MFT hamiltonian is invariant by a $\sigma \leftrightarrow \overline{\sigma}$ exchange, thus all measurable physical quantities must be. What is unmeasurable and has no effect on Physics is free to be σ-dependent, as for the sign of $\mathrm{Re}\{\Delta_{\mathbf{k}\sigma}\}$. Since all it's relevant of both the complex components of the gap function is their magnitude, we can safely treat σ as a pure degeneracy. This gives a simple equation for the last HFP,

$$\begin{aligned} m &= \frac{1}{L_x L_y} \sum_{\mathbf{k}\in\mathrm{MBZ}} \langle \hat{s}^x_{\mathbf{k}\uparrow} \rangle \\ &= -\frac{1}{2L_x L_y} \sum_{\mathbf{k}\in\mathrm{MBZ}} \frac{\mathrm{Re}\{\tilde{\Delta}_{\mathbf{k}}\}}{\tilde{E}_{\mathbf{k}}} \tilde{\mathrm{Th}}^{(-)}_{\mathbf{k}}[\beta, \tilde{\mu}] \end{aligned} \tag{2.33}$$

[To be continued...]

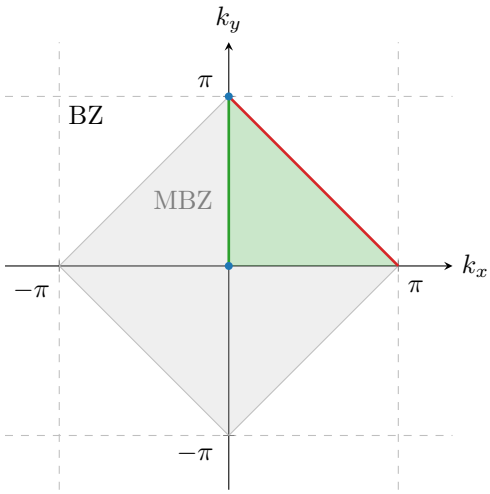
2.5 Stability of the AF phase

The discussion above is formally correct and self-consistent, thus a HF algorithm can be sketched to extract the HFPs. However we cannot proceed without considering the instability effects of Sec. **??**. As is known (and we would have liked to notice earlier) the AF-SSB at wavevector $\boldsymbol{\pi}$ we are studying here, responsible of halving the BZ down to the MBZ, can produce a stable phase on pure Hubbard lattices only at half filling [**singh1990collective**]. For the EHM the situation is even worse than that: repeating the simple Linear Reponse Theory (LRT) computations of Sec. **??** and mimicking Eq. (**??**), we get a proper response function

$$\tilde{\chi}_{\hat{\mathbf{S}}^-\hat{\mathbf{S}}^+}(\mathbf{k},\omega) = \frac{\chi_{\hat{\mathbf{S}}^-\hat{\mathbf{S}}^+}(\mathbf{k},\omega)}{1 - [U - 2V(\cos k_x + \cos k_y)]\,\chi_{\hat{\mathbf{S}}^-\hat{\mathbf{S}}^+}(\mathbf{k},\omega)} \tag{2.34}$$

[Double check the red part and conclude this paragraph...].

As is well documented in literature [**wietek2021stripes**], AF can establish in a wide range of spatial structures with particular shapes, for example stripes. In order to extract the precise AF phase, we should



Color code	$w(\mathbf{k})$
White	0
Gray	0
Blue	1
Red	2
Green	4

Figure 2.3 Graphic rendition of s^*-wave symmetry over the MBZ employed to optimize calculations. The right-side table indicates, for any point in BZ, the relative weight assigned to calculation.

1. Choose an initializer wavevector, say $\mathbf{k} = \boldsymbol{\pi}$;
2. Solve self-consistently for the HFPs at $\mathbf{k}$;
3. Compute the free energy F as a functional of the wavevector around $\mathbf{k}$;
4. Look for gradient descent of the free energy, $\boldsymbol{\nabla} F < 0$;
5. If the gradient descent is found, update $\mathbf{k} \to \mathbf{k} + \delta\mathbf{k}$ and repeat from step 2, otherwise halt.

Of course the sketched procedure can lead to a lot of complications, first the fact that at incommensurate wavevector $\mathbf{k} \neq \boldsymbol{\pi}$ the unit cell is not as simple as a pair of sites, but can become much larger requiring more refined computational solutions.

In this text we will show the self consistent result for AF-SSB at wavevector $\boldsymbol{\pi}$ even at finite doping as a first step in the above sketched procedure. The presented results are self-consistent, but unfortunately deeply unstable and unphysical at finite doping, a situation where an incommensurate and insulating AF phase is preferred to this commensurate metallic counterpart. Nevertheless we present them anyway.

2.6 HF algorithm, computational strategy and results

This section is devoted to the discussion of the self-consistent results we obtained running an iterative HF algorithm in the parameters space. As said earlier, not all presented results are in fact stable; however we decided to show them as sketch of the first step in a general self-consistent algorithm in search of the stablest AF structure.

2.6.1 Preliminary symmetry considerations

A very important feature when dealing with optimization is recognizing the complete s^* symmetry of the model, and the HFP parameters with relative self-consistency equations. Consider Eqns. (**??**), (**??**) and (**??**): in all of three the term inside the sum exhibits s^*-wave symmetry and is summed over the MBZ. Then there is no need of sweeping the entirety of the MBZ, it suffices to sweep just one fourth and just multiply coherently the result.

A little care is necessary when dealing with borders, due to MBZ nesting. Consider in fact the MBZ rhombus, sketched in gray in Fig. 2.3: due to periodicity, two of its four boundaries must be excluded from computation in order to avoid redundancy. Let those be the lower boundaries,

$$k_y = |k_x| - \pi$$

Due to periodicity, the remaining upper boundaries lead to identical results, thus we may compute just one of them and multiply the result by 2. In doing this, we need to avoid the edges: due to nesting of the MBZ, the three points

$$(0, \pi) \qquad (-\pi, 0) \qquad (\pi, 0)$$

are the same point. Thus these need to be considered once. Looking to the bulk of the rhombus, due to s^* periodicity it suffices to integrate over one quarter (considering just one internal border) and multiply by 4. In doing this, the origin $(0, 0)$ must be counted only once. Fig. 2.3 summarizes this argument: if we assign to any given point $\mathbf{k} \in \mathrm{BZ}$ a weight function $w(\mathbf{k})$ defined as in figure, in particular avoiding computations over white and gray points, we get an optimized integral over the entire MBZ saving *circa* 75% of runtime.